Who Benefits from Benefits?
Municipal Consequences of Federal Cash Transfers

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Abstract

Centralized welfare programs limit local discretion over eligibility and payments, yet access often depends on municipal bureaucracies. We study whether local politicians obtain electoral returns from a federally designed cash transfer when their remaining discretion lies in registration. In Brazil's *Bolsa Família*, a 2009 reform unexpectedly changed the formula assigning municipal beneficiary quotas, and a later federal revision opened additional slots during the same mayoral term. We measure the cumulative authorization change relative to the allocation implied by the previous rule. In the 2,078 municipalities where the mayor elected in 2008 sought reelection in 2012, a one-standard-deviation quota shock created 171.50 additional authorized slots per 10,000 residents and induced 62.14 additional beneficiary families per 10,000 residents. The 2SLS estimate implies that 100 additional beneficiary families per 10,000 residents raise vote share by 1.70 percentage points. Enrollment expanded both with and without a pre-existing local welfare center (*CRAS*), but we find a clear electoral response only in *CRAS* municipalities. Within that group, the vote-share response is larger where offices report active outreach and more intensive service to *Bolsa Família* families. We interpret these patterns as scope conditions for the local implementation channel. Centralizing eligibility and payment does not eliminate local political returns when program access still depends on municipal implementation.

Keywords: cash transfers; electoral returns; state capacity; *Bolsa Família*.
JEL Codes: D72, H53, H75, I38, O17.

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1 Introduction

Across the developing world, cash transfer programs have become a leading tool for reducing poverty while insulating redistribution from local political discretion. By standardizing eligibility rules, fixing benefit values by federal decree, and processing payments through centralized banking systems, these programs are designed to reduce the scope for patronage and discretionary allocation (Fiszbein and Schady, 2009). Yet access often depends on local bureaucracies controlled by elected officials. Can local politicians benefit electorally from policies designed and financed by another level of government? This is the question we address.

We study this question in the context of Brazil's *Bolsa Família*, one of the largest conditional cash transfer programs in the world.[1] Although eligibility rules, benefit size, and payments are determined by the federal government, municipalities remain responsible for identifying and registering eligible households, mostly through local welfare centers known as *Centros de Referência de Assistência Social* (*CRAS*). This institutional arrangement creates a narrow but important margin of local discretion: while mayors cannot alter transfer amounts, redirect funds, or withhold payments, they control the administrative pipeline through which households may gain access to the program.

This margin depended on a registration process that required active municipal effort. Identifying potentially eligible families meant locating them, conducting household interviews, and collecting documentation; municipal authorities controlled the local registration apparatus, which gave them influence over the pace, intensity, and reach of enrollment efforts.[2] In practice, two municipalities facing similar levels of underlying need could generate very different realized coverage depending on how actively the local administration pursued registration.

Our empirical strategy starts from a 2009 reform that revised the formula used to compute municipal *Bolsa Família* quotas. The program operates through federally determined municipal quotas, which set a ceiling on the number of families that can be served in each municipality.[3] A mayor elected in October 2008 could expect the

[1] By June 2006, *Bolsa Família* covered 11.10 million families, approximately 46 million people, or about 25% of the Brazilian population. By June 2015, the program supported 13.80 million families, with an average monthly benefit of R$167.15 and a budget equivalent to 0.50% of GDP.

[2] Hellmann (2015) describes enrollment as a municipal responsibility. He reports that as of September 2014, 77% of registrations were made without a home visit. Municipalities were responsible for preparing the local structure for registration, training personnel, promoting awareness campaigns, and designing *active search* strategies to reach eligible families outside the existing registry.

[3] Lindert et al. (2007) explains that *Bolsa Família* maintained municipal program quotas based on local

next adjustment to follow the previous allocation rule. Instead, the federal government changed that rule in 2009 and revised municipal quotas again in 2012.[4] These revisions expanded the ceiling, not the registry. Families who were already registered could enter when new slots opened, while filling the remaining slots depended on local governments' effort to identify, locate, and register eligible families not yet in the federal database.

We combine administrative data on quotas, realized beneficiary counts, and local social-assistance infrastructure with municipal electoral data from the subsequent election. The construction of the 2009 benchmark follows Gerard et al. (2026): we apply the previous allocation rule to the information available before the reform. Our main measure subtracts that benchmark from the official 2012 quota, capturing the cumulative authorization that opened during the mayor's term. We use this quota change to instrument realized beneficiary expansion.

Our findings trace the reform along four stages: the relationship between federal authorization and realized enrollment; the electoral response to the authorization shock; the 2SLS estimate that combines the two; and the institutional conditions under which these patterns emerge.

The first stage quantifies how federal authorization translated into realized enrollment. A one-standard-deviation increase in the quota shock corresponds to 171.50 additional authorized beneficiary-family slots per 10,000 residents. It raises realized beneficiary expansion by 0.34 standard deviations, equivalent to 62.14 additional beneficiary families per 10,000 residents.

Once federal authorization is shown to translate into realized enrollment, we estimate the electoral return to expanding the ceiling itself, before separating out how much of that ceiling municipalities chose to fill. For the same one-standard-deviation quota shock, the reduced-form intention-to-treat estimate is 1.06 percentage points on the incumbent's vote share. This is the electoral consequence of additional federal authorization, integrating over the local enrollment response documented in the first stage.

Scaling the intention-to-treat effect by the first stage gives the electoral response

poverty estimates. Original municipal allocations were constructed by combining the 2001 PNAD with the 2000 Census, generating an initial national target of 11.20 million families when the program was created in 2003.

[4] Hellmann (2015) notes that *Bolsa Família* administered municipal quotas using estimates of poor families developed in partnership with IBGE and reports that, in 2009, the federal government refined the measurement of the target population by incorporating methodologies to capture income volatility among vulnerable households. These studies formed a major expansion of coverage, incorporating more than 1.60 million additional families.

to realized beneficiary expansion. A one-standard-deviation increase in instrumented enrollment, about 181.36 additional beneficiary families per 10,000 residents, raises incumbent vote share by 3.08 percentage points. Equivalently, an increase of 100 beneficiary families per 10,000 residents raises vote share by 1.70 percentage points. The estimate is similar across alternative quota definitions, outcome definitions, and corrections for selection into the rerunning sample.

These returns differ across the institutional layer that converts authorization into delivery. Of the 2,078 municipalities in the estimation sample, 1,394 had a pre-existing local welfare center in 2008 and 684 did not. Realized enrollment expands in both groups: the first-stage coefficients are 0.34 with a *CRAS* and 0.35 without one. The divergence is electoral. The corresponding 2SLS estimate is 4.10 in the *CRAS* group; outside it, the confidence set includes zero. These estimates locate a clearer electoral response in municipalities with established offices, although balancing observed municipal characteristics substantially reduces that contrast.

Within these municipalities, we next examine two measures of how the local welfare apparatus operated. The first records whether municipal teams proactively searched for potentially eligible families (active outreach) rather than waiting for them to appear at a registration point. The first stage remains positive both with and without this practice, while the reduced-form vote-share point estimate is 1.62 percentage points with active outreach and 0.25 without it. The second measure records whether the *CRAS* reported serving *Bolsa Família* families through PAIF (*Serviço de Proteção e Atendimento Integral à Família*), a continuing family-support service provided through the social-assistance system. The corresponding reduced-form estimates are 1.62 percentage points where at least one *Bolsa Família* family was served through PAIF and 0.69 where none was reported. The PAIF contrast becomes sharper when we consider service volume: among municipalities reporting at least one family served through PAIF, the reduced form is 4.25 above the median number of families served and indistinguishable from zero below it. Because both operational measures predate the quota shock but were not randomly assigned, we interpret these results as scope patterns. They motivate a contact interpretation, but subsequent tests of operating duration and reported list-use frequency do not establish a consistent electoral gradient. Comparisons across office-opening cohorts are descriptive, and the evidence leaves the specific CRAS mechanism unresolved.

These findings advance three related literatures. We speak first to the political economy of cash transfer programs. Studies of social transfers find gains in government support and, in some settings, incumbent voting, although electoral results vary across de-

signs and contexts (Manacorda et al., 2011; De La O, 2013; Imai et al., 2020; Frey, 2019). Much of this literature studies settings in which local discretion is deliberately minimized, relying on centrally determined thresholds, randomized assignment, or other exogenous variation to identify electoral effects. We complement this work by studying a setting in which the program architecture is centralized but the last mile of delivery is not. Our findings suggest that centralizing transfer rules and payment flows is not sufficient to depoliticize redistribution when access to the program depends on a locally controlled registration bureaucracy. Even when mayors cannot alter eligibility criteria or benefit levels, control over the administrative pipeline through which households gain program access can still generate measurable electoral returns.

Our paper is especially closely related to Frey (2019), but the two designs study electoral returns under different implementation regimes: Frey's intervention leaves less room for mayoral involvement, whereas ours depends directly on municipal administration. Frey exploits a discontinuity in federal funding for the *Family Health Program*, a primary-care program in which teams follow a federally specified protocol of routine home visits to poor households. His information channel is embedded in those visits: the teams disseminate *Bolsa Família* eligibility information as part of their routine work, making enrollment less dependent on a separate effort by the mayor's welfare administration. We study the complementary margin. In our setting, converting expanded federal authorization into realized enrollment requires municipal welfare staff to identify, locate, and register eligible families. The designs also draw variation from different parts of the municipal distribution. Frey's threshold design is concentrated among small, low-development municipalities near a funding threshold, whereas the quota reform created more room for expansion in larger municipalities that had not yet absorbed earlier increases. Within his sample, coverage growth through a federally structured outreach channel reduces incumbency advantage and shifts the composition of challengers toward less clientelistic parties. Within ours, coverage growth through a locally managed registration channel produces positive returns for the incumbent.

Second, we contribute to the literature on political accountability under shared governance. A central question in this literature is whether voters can attribute responsibility when multiple levels of government jointly produce policy outcomes (Przeworski et al., 1999; Besley, 2006). Ferraz and Finan (2008) show that voters punish corrupt mayors when randomly assigned audits reveal misuse of federal funds, especially where local media facilitate information transmission. Avis et al. (2018) show that audits also reduce corruption through a legal disciplining channel. Our setting differs because the

relevant outcome is positive delivery rather than malfeasance. The results are consistent with voters giving local incumbents some credit for the expansion of a federally funded program when access still depends on municipal action. We do not observe voters' beliefs directly, but the pattern fits an attribution channel in a setting of shared responsibility.

Third, we contribute to the literature on state capacity and the bureaucratic foundations of policy delivery. Besley and Persson (2009, 2011) establish state capacity as a precondition for credible policy delivery, and Finan et al. (2017) survey the personnel and bureaucratic infrastructure through which public programs are implemented. Our findings distinguish enrollment pass-through from political conversion. Municipalities both with and without a pre-existing *CRAS* translate the quota shock into additional beneficiaries. The initial electoral contrast favors municipalities where the delivery apparatus was already present, but is sensitive to their observed composition. This distinguishes the evidence for local electoral returns from the more demanding claim that welfare offices themselves produce those returns.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting of *Bolsa Família*, the quota system, the federal revisions, and the data sources. Section 3 presents the impacts of the reform and summarizes the identification strategy, while the full empirical strategy appears in Online Appendix A and the figures and tables are collected after the references. Section 4 examines local implementation as a possible mechanism, evaluates the institutional comparisons, and discusses alternative explanations. Section 5 concludes.

2 Institutional Background and Data

2.1 *Bolsa Família* Program

Program Design. The federal government finances *Bolsa Família*, defines who qualifies, and pays eligible families. Municipal administrations help families enter the national registry on which federal selection depends. This division gives families a reason to encounter municipal staff while seeking a federally funded transfer, and gives mayors a possible electoral return from facilitating access.

Launched in 2004, *Bolsa Família* consolidated earlier transfer programs. In 2009, families with monthly per capita income at or below R$70 qualified for a basic benefit and age-related variable benefits; those with income above R$70 and at or below

R$140 qualified for variable benefits. Health and education conditions applied to the relevant family members. Table 1 gives the amounts, limits, and changes in the eligibility thresholds. The statutory income ranges matter for interpreting which families entered the registry after quotas expanded.

Federal Rules and Local Responsibility. *Bolsa Família* combined centralized rules and payments with decentralized administration. The federal government, through the *Ministério do Desenvolvimento Social* (MDS), set eligibility criteria and benefit values, used *Cadastro Único* records to identify families eligible for payment, and transferred benefits directly through the state-owned bank *Caixa Econômica Federal*. State governments coordinated implementation across social assistance, education, and health and provided technical support to municipalities, but neither selected beneficiaries nor disbursed transfers.[5]

Municipal governments operated the program locally. They identified low-income families, collected and updated their registry information, conducted household visits, monitored compliance with education and health conditions, and coordinated the municipal agencies involved in implementation. Mayors therefore could not change eligibility thresholds or benefit values, redirect federal transfers, or guarantee that a registered family would receive a benefit. Their administrations could, however, affect how intensively potential beneficiaries were sought, how quickly applications were processed, and how completely eligible families became visible to the federal system. This separation between federal benefit rules and local implementation is the institutional margin central to our analysis.

Municipal Registration. The *Cadastro Único* (*CadÚnico*) is a national registry of low-income families and the administrative gateway to *Bolsa Família* and many other social programs. It covered families with per capita income below half the minimum wage (R$232.50 in 2009), well above the program's own eligibility thresholds. Registration was therefore necessary but not sufficient for benefit receipt, and the registered population was substantially larger than the beneficiary population. Under Decree No. 6,135/2007, the MDS defined and supervised the national registry while participating municipalities collected family information using a standardized form.

Registration required a face-to-face interview covering household composition,

[5]The division of responsibilities during the study period was governed principally by Law No. 10,836/2004, Decree No. 5,209/2004, and MDS Ordinance No. 246/2005. The ordinance formalized municipal adherence to the program and specified federal and municipal commitments.

housing, employment, and income. A family could visit a municipal registration post, or a team could travel to its neighborhood or rural community. The municipal task involved locating families, conducting interviews, updating records, and coordinating referrals. New registration and benefit receipt are different events: a newly registered family might not receive a transfer, and an already registered family might fill a newly authorized place.

The Local Welfare Office. A *Centro de Referência de Assistência Social* (*CRAS*) is a territorial public welfare office within the *Sistema Único de Assistência Social* (SUAS). Its staff receive families, provide basic social-assistance services, and connect them with other parts of the protection network. The *Programa de Atenção Integral à Família* (PAIF) supplied continuing family support through these offices. Operating a *CRAS* therefore requires premises, staff, territorial knowledge, management, and coordination with municipal registration teams. The office may host registration or help a family reach it; a municipality may also register families elsewhere. This distinction lets us ask whether an existing welfare interface changes the political return to expansion even where enrollment can grow without one.[6]

Why Offices Were Unevenly Distributed. Pre-reform *CRAS* presence reflects both national allocation rules and municipal implementation. Federal cofinancing flowed through the social-assistance fund and supported PAIF services rather than household cash payments. The 2006 expansion required municipalities to qualify for basic or full SUAS management; its allocation ranking combined poverty, fiscal capacity, previous federal transfers, and management indicators. Population and state groups, as well as priority for first coverage, also shaped offers. The 2008 expansion added an SUAS ranking and other priority categories. Municipalities then had to accept the funding, demonstrate implementation capacity, and provide the service. An authorization or funding offer is consequently not proof that an office was operating before the reform.[7]

Municipalities with a recorded office may therefore differ in poverty, size, fiscal resources, administrative organization, and earlier welfare provision. We use pre-reform office presence to describe where the electoral response appears; its allocation history

[6]The MDS, Portaria No. 385/2005 and Portaria No. 225/2006, and CIT Resolution No. 3/2008 specify the financing and expansion rules. The MDS, *Orientações Técnicas: Centro de Referência de Assistência Social* (2009), pp. 15–17 and 25, distinguishes territorial services from municipal registry management. The 2009 *Censo SUAS* counted 5,798 offices in 4,329 municipalities.

[7]See Portaria No. 225/2006, arts. 2–3, and CIT Resolution No. 3/2008, arts. 1–2 and implementation stages. These rules describe national cofinancing, not the complete construction or financing history of each office.

makes a causal interpretation of office presence demanding.

2.2 Institutional Details of the Quota Reform

Quotas and Enrollment. Eligibility alone does not guarantee benefit receipt. Access to *Bolsa Família* depends on the availability of municipal quotas: caps on the number of beneficiaries per municipality, defined nationally and revised approximately every three years based on poverty estimates derived from census and survey data. These quotas function as official upper bounds on municipal enrollment, but the constraint is not always immediately binding. In practice, municipalities frequently operate near their ceiling, oscillating modestly above or below it from month to month before the federal priority mechanism reallocates unused slots.

Priority and Local Effort. The priority index considers both quota utilization and unmet registered demand: municipalities with low beneficiary-to-quota ratios and few eligible-but-unserved families risk seeing unused slots reallocated to places with stronger registered demand. The quota system therefore creates pressure to use the available slots. A quota increase is not a benefit guarantee, but it opens federally authorized room that municipalities must fill by identifying, registering, and updating eligible households. Effective coverage thus depends on local administrative initiative: political will and organizational capacity determine how much of the authorized ceiling becomes actual benefit receipt.

Quota Reform. In 2009, the federal government revised the methodology used to compute municipal quotas, producing a heterogeneous reallocation of authorized beneficiary slots across municipalities. The previous approach, in place since 2006, distributed slots in proportion to each municipality's estimated number of poor families from the 2000 Census, adjusted for intercensal population growth. The revised 2009 methodology accounts for income volatility among families near the eligibility line, a feature of transient poverty that earlier projections systematically undercount (Gerard et al., 2026).

Reallocation Shock. Alongside the methodological update, municipal quotas were scaled by a common factor of 1.18 to account for income volatility near the poverty threshold. Applied to unequal baseline exposures, even a common multiplier can contribute to differences in quota changes across municipalities. Our instrument compares

actual authorization with the allocation implied by the earlier rule; its identifying assumption concerns the remaining variation after the stated baseline controls, not the federal origin of the formula alone. The 2009 expansion added nearly 1.60 million authorized places. An already registered family could fill a place without a new registration, while reaching a family outside the registry required additional municipal work.

The subsequent municipal reallocation also preceded the 2012 election. The federal government had raised its national coverage target to 13.8 million families in 2011, so another national expansion was public information. In May 2012, however, MDS announced how that target had been distributed across municipalities using the 2010 Census sample released in April. Our design uses the cumulative municipal authorization available before voting because both the 2009 reform and the 2012 reallocation could affect coverage and electoral outcomes during the mayoral term. The relevant claim is about the formula-driven distribution of local changes, not surprise about the national target.[8]

2.3 Data

Our empirical analysis combines five sources of data: administrative records on *Bolsa Família*, family-level records from the *Cadastro Único*, electoral returns, census and household-survey data, and administrative records on local social-assistance infrastructure.

***Bolsa Família* Administrative Records.** We begin with monthly administrative records from the *Ministério do Desenvolvimento Social* on *Bolsa Família* quotas, realized coverage, and total transfers from 2006 to 2012. These data span the pre-reform baseline, the 2009 reallocation, and the post-reform expansion window through the 2012 election, and allow us to measure both authorized and realized program expansion.

***Cadastro Único*.** We also use family-level records from the 2012 *Cadastro Único*, provided by the *Ministério do Desenvolvimento Social*. The file covers families recorded in the registry in 2012, including those registered in earlier years. It reports their municipality of residence, per-capita family income, family characteristics, and dates of first registration and last update. The first-registration date identifies entry into the reg-

[8] MDS, *Informe Bolsa Família* No. 318, May 17, 2012, pp. 1–2. The bulletin identifies the 2011 national target, April 2012 Census-sample release, municipal allocation, and the continuing municipal role in locating and registering vulnerable families.

istry separately from subsequent updates. We use it to construct registration cohorts, including families first registered before the reform.

Snapshot and Updating. Income and family characteristics reflect the information on file in 2012. During the study period, registry information required updating or revalidation every two years.[9] A family first registered in 2007 can therefore be assigned to its registration cohort using the 2012 file. Its recorded income, however, describes its circumstances as last reported by 2012. The cohort comparisons describe families observed in that snapshot, grouped by first-registration date and income recorded in 2012. Families absent from the snapshot and their earlier income histories remain outside this exercise.

Electoral Records. We then use electoral records from the *Tribunal Superior Eleitoral* (*TSE*) for the 2008 and 2012 municipal elections. These data provide candidate-level vote totals, turnout, and detailed candidate characteristics, including education, prior occupation, and party affiliation.

Census and Household Surveys. To reconstruct counterfactual municipal quotas under the pre-reform allocation rule, we combine the 2000 Census from the *Instituto Brasileiro de Geografia e Estatística* (*IBGE*) with poverty estimates from the 2006 *Pesquisa Nacional por Amostra de Domicílios* (*PNAD*). These are the inputs used to approximate the quota formula that would have prevailed without the 2009 reform.

Censo SUAS. Finally, we use administrative records from the *Censo SUAS* covering *CRAS* units in 2008 and 2009. These data describe local infrastructure, active outreach, and recorded numbers of beneficiary families served. We use the family count as a provisional proxy for PAIF activity, subject to the measurement limits discussed in Section 4.3. To examine office duration and network expansion, we also harmonize the annual censuses from 2007 through 2012. Absence from a census is treated as missing information rather than evidence that an office closed.

[9] Article 7 of Decree No. 6,135 of June 26, 2007 established the two-year validity of registry information.

2.4 Sample Construction

Municipal Panel and Data Availability. Our electoral analysis follows municipalities in the 2008 and 2012 elections. We combine the sources using municipal identifiers and require both election years, valid quota and coverage measures, and the covariates needed for the common specification. Missing records identify gaps in the measures needed for estimation; they do not establish that a municipality lacked access to *Bolsa Família*. Authorization and realized expansion are measured in family counts per 10,000 residents using fixed 2006 population.

Electoral Sample and Record Quality. We follow the mayor elected in 2008 and measure that mayor's vote share in each election. The electoral analysis requires that this mayor contest the 2012 election, that both races have usable vote records, and that the candidate can be linked unambiguously across elections. The resulting estimand concerns incumbent candidates who sought another term.

Final Estimation Sample and Selection. Complete information on municipal transfers excludes 24 municipalities, and missing mayor characteristics exclude another 3. The final sample contains 2,078 municipalities and 4,156 municipality-year observations. We use this common sample across the main specifications. Its construction also requires the variables used in the alternative windows and mechanism comparisons. Table 2 compares it with a broader descriptive sample of 4,976 municipalities. The selection diagnostic uses a separate complete-case universe of 5,492 municipalities before conditioning on candidacy.

Selection into Rerunning. Selection matters because the reform could affect whether the incumbent runs again. We examine the association of the quota shock with rerunning in a broader sample and reweight the electoral regression by observed rerunning probabilities. Neither exercise rules out selection on unobserved determinants.

2.5 Descriptive Statistics

Municipality and Mayor Characteristics. Table 2 describes municipalities and their elected mayors in the broader comparison sample and the estimation sample. The 2008 winner's vote share averages 54.97 percent in the former and 52.92 percent in the latter. Predetermined municipal characteristics, including urbanization, poverty, income,

and inequality, have similar means across the two samples. Mayor characteristics differ more, including prior occupation and age. The estimation sample describes incumbents who contest another election, so these differences help delimit the population studied. Similar municipal averages alone do not establish that selection is unrelated to electoral outcomes.

Program Coverage and Expansion. Municipalities in the estimation sample had an average of 879.87 beneficiary families per 10,000 residents in 2008. Realized coverage rises by 203.50 families per 10,000 residents on average through 2012, with a median increase of 163.15. The realized measures use twelve-month averages of monthly beneficiary-family counts. Rates use municipal population in 2006 as a fixed denominator, so changes reflect enrollment rather than contemporaneous population growth. Authorized and realized expansion vary across municipalities and are right-skewed. We express regression coefficients per one analysis-weighted standard deviation to make their scale interpretable and translate the estimates into families per 10,000 residents. These measures count families receiving benefits, whereas the *Cadastro Único* cohort exercise describes registered families using the 2012 snapshot.

Local Social-Assistance Infrastructure. The estimation sample includes 1,394 municipalities with a pre-existing *CRAS* and 684 without one. Municipalities with a *CRAS* account for 67.08 percent of the sample. The infrastructure measures also distinguish whether these units conduct active outreach and provide family-assistance services. These characteristics describe the local capacity through which eligible families may enter the registry and obtain benefits. Section 4 examines whether the electoral response to expansion differs across these settings. That comparison conditions on existing infrastructure and does not identify the effect of installing a *CRAS*.

The figures and tables discussed below are collected after the references.

3 Impact of the 2009 *Bolsa Família* Reform

Can mayors gain electoral support from the expansion of a federal transfer program when municipalities are only responsible for finding and registering eligible families? Federal policy determines how many families may receive benefits and the transfers sizes, but municipal staff help turn that authorization into access. We compare municipalities whose authorized capacity expanded by different amounts and trace the cor-

responding changes in registration and enrollment. Using the formula-driven change in authorized capacity as an instrument for registrations and realized enrollment, we estimate its effect on electoral outcomes.

3.1 Policy-Induced Variation in Authorization and Coverage

Authorization. We begin by documenting the change in federal authorization that underlies our design. In 2009, the Ministry of Social Development revised the formula for calculating municipal *Bolsa Família* quotas, then updated the quota estimates before the 2012 election using newly released data from the 2010 Census. We compare each municipality's official 2012 quota with the quota it would have received in 2009 under the earlier allocation rule. This difference measures the additional federal authorization available during the mayoral term and provides our instrument for registrations in *Cadastro Único* and realized enrollment growth. Figure 1 uses registration cohorts to show that roughly one-third of municipalities had fewer potentially eligible registrations than authorized slots in 2008, leaving scope for new registrations; the share fell below one-tenth by 2012. Figure 2 shows how the reform opened a gap between authorized quotas and realized enrollment. Figure 3 compares unused authorization under the official 2012 and counterfactual quotas.

Registrations. Municipalities responded to larger quotas by registering more families in *Cadastro Único*. Under the complete specification, a one-standard-deviation quota increase is associated with 48.47 additional registrations per 10,000 residents during the mayoral term (Table 3). The response differs by recorded family income: 43.60 of the additional registrations per 10,000 residents are among families with 2012 income at or below R$140 per capita, including 34.32 at or below R$70. The estimate for families above R$140 is 4.87 and statistically imprecise, so the evidence points to an increase concentrated among potential beneficiaries, without evidence of increased registrations above the potential-beneficiary threshold. Figure 4 plots cumulative registration cohorts by reported income, distinguishing families at or below the potential-eligibility threshold from higher-income families.

Realized Enrollment. Registration creates a path to *Bolsa Família*, but a place in *Cadastro Único* does not itself confer a benefit. To assess whether expanded authorization and registration translated into *Bolsa Família* benefit receipt, we compare the twelve-month average number of beneficiary families in 2008, before the mayoral term, with the corre-

sponding average in 2012, the election year at its end. Under the complete specification, a one-standard-deviation quota increase is associated with 62.14 additional beneficiary families per 10,000 residents (Table 4). Figure 5 compares quotas and realized enrollment at the beginning and end of the mayoral term. Because the outcome is a change in beneficiary-family stocks, it cannot distinguish how many of the added beneficiaries were registered during the term rather than before it.

3.2 Identification Strategy

We estimate the electoral effects of the 2009 expansion of *Bolsa Família* using a two-period difference-in-differences design that compares changes in electoral outcomes from 2008 to 2012 across municipalities exposed to different increases in federal quota authorization. Because registrations and realized enrollment may also reflect municipal choices, we use the authorization increase to isolate federally induced beneficiary growth. Formally, we estimate

$$\Delta \mathrm{B}_{\mathrm{m}} = \pi_0 + \pi_1 \mathrm{Z}_{\mathrm{m}} + \mathbf{X}'_{\mathrm{m},0}\boldsymbol{\pi} + \eta_{\mathrm{m}}, \tag{1}$$

$$\Delta \mathrm{Y}_{\mathrm{m}} = \beta_0 + \beta_1 \widehat{\Delta \mathrm{B}}_{\mathrm{m}} + \mathbf{X}'_{\mathrm{m},0}\boldsymbol{\beta} + \varepsilon_{\mathrm{m}}, \tag{2}$$

where $\Delta \mathrm{Y}_{\mathrm{m}}$ is the 2008–2012 change in an electoral outcome for municipality m. The variable $\Delta \mathrm{B}_{\mathrm{m}}$ is the change in program coverage over the same period. We examine both registrations and realized enrollment as responses to authorization in the first equation; the second equation estimates its electoral effect. The instrument Z_{m} measures the increase in authorized quotas relative to the pre-reform allocation rule. The vector $\mathbf{X}_{\mathrm{m},0}$ contains the full set of predetermined controls used in each specification.

Counterfactual Quotas We construct the counterfactual quota by applying the pre-reform allocation rule to information available before the reform, following Gerard et al. (2026):

$$\mathrm{CounterfactualQuota}_{\mathrm{m}}^{2009} = \frac{\mathrm{Poor}_{\mathrm{m}}^{2000}\,\mathrm{Growth}_{\mathrm{m}}^{2000,\,2006}}{\sum_{\mathrm{k}\in\mathrm{s}} \mathrm{Poor}_{\mathrm{k}}^{2000}\,\mathrm{Growth}_{\mathrm{k}}^{2000,\,2006}} \times \mathrm{Poor}_{\mathrm{s}}^{2006} \times 1.18. \tag{3}$$

where $\mathrm{Poor}_{\mathrm{m}}^{2000}$ is the number of poor families in municipality m in the 2000 Census, $\mathrm{Growth}_{\mathrm{m}}^{2000,\,2006}$ is its estimated population growth factor through 2006, and $\mathrm{Poor}_{\mathrm{s}}^{2006}$ is the 2006 PNAD survey estimate of poor families in state s, which contains municipality m. The factor 1.18 incorporates the nationwide increase in authorized slots that accompanied the 2009 revision.

Instrument We then define the instrument as

$$Z_m = \frac{\text{OfficialQuota}_m^{2012} - \text{CounterfactualQuota}_m^{2009}}{\text{Population}_m^{2006}} \times 10{,}000. \tag{4}$$

where $\text{OfficialQuota}_m^{2012}$ is the quota assigned to municipality m before the 2012 election, and $\text{Population}_m^{2006}$ is the population estimate available before the 2008 election.

Our instrument extends the comparison in Gerard et al. (2026) from 2009 to 2012 to capture the authorization available over the mayor's full term. Following their application of the recentering logic in Borusyak and Hull (2023), we subtract the quota implied by the pre-reform rule from the official quota. This removes the allocation the municipality would have received under the old rule, leaving the additional authorization generated by the federal changes. We treat this cumulative change as plausibly exogenous to the mayor elected in 2008: the federal government changed the quota allocation formula in 2009 and updated municipal quotas in 2012, and the municipality's eventual allocation was neither chosen by the mayor nor known at that election.

Validity of the Design. For the electoral comparison to identify an effect, municipalities receiving different shocks must have followed comparable electoral trends in the absence of the expansion, conditional on predetermined characteristics. The instrumental-variable interpretation further requires that authorization affect electoral outcomes only through beneficiary growth and that higher authorization not induce lower enrollment. The results below examine the observable implications of these conditions and the sensitivity of the estimates to alternative specifications.

3.3 Electoral Responses

In this section, we estimate the causal effects of the the *Bolsa Família* expansion on a set of electoral outcomes. We show that, quota-induced registration and beneficiary growth increased the vote share of mayors seeking reelection, while the evidence do not establish effects on their probability of winning or on turnout. The positive mayoral return contrasts with the negative local-incumbency response in Frey (2019) and echoes Labonne (2013)'s finding that local incumbents gained votes where a conditional cash transfer reached more communities. Taken together, the results are consistent with municipal involvement in delivering a federal benefit creating an opportunity for mayors to gain electoral support.

Authorization and Incumbent Vote Share. The authorization comparison asks whether mayors receive more votes where the federal government permits greater expansion. The preferred reduced-form estimate is 1.06 percentage points per standard deviation of the quota shock (Table 5); the lean-column estimate is 1.86. This comparison captures the electoral response to authorization, including any response that operates before families actually receive benefits.

Realized Enrollment and Incumbent Vote Share. In the population-weighted specification, instrumenting enrollment with authorization gives 3.08 percentage points of incumbent vote share per one weighted standard deviation of induced beneficiary-family expansion ([0.74, 6.40]; Table 6). The Anderson–Rubin set excludes zero in the preferred column. Rescaled, 100 additional beneficiary families per 10,000 residents correspond to 1.70 percentage points ([0.41, 3.53]). The median observed expansion was 163.15 families per 10,000 residents; applying the pooled coefficient to that dose gives approximately 2.77 percentage points. This is a scale illustration, not an estimated effect for a particular median municipality, and not a claim that all its observed expansion was caused by the reform. The median incumbent lost 3.03 percentage points of vote share. These comparisons make the electoral magnitude visible while leaving individual voter behavior unresolved.

Table 7 rescales the same electoral reduced form by quota-induced first registrations. That comparison measures a different expansion margin; it cannot separate the electoral return to registration from the return to subsequent benefit receipt.

Winning Conditional on Rerunning. The winning outcome records whether the mayor elected in 2008 won in 2012 among mayors who appeared on the 2012 ballot. Its point estimate is positive under the complete specification, but the Anderson–Rubin confidence set includes zero (Table 6). It is not the unconditional probability that a mayor was reelected: the design conditions on the decision to run again.

Turnout. Turnout provides a separate margin of electoral response. We do not find a clear turnout response to local program expansion; the confidence set includes zero. The preferred response to authorization is -0.33 percentage points, and the response to instrumented enrollment is -0.96 ([-2.46, 0.21]). Tables 5 and 6 report all three outcomes under the same two control sets. These aggregate outcomes cannot reveal whether an individual switched candidates or an opponent lost votes.

3.4 Credibility and Limits

The comparison remains sensitive to how the national rule maps municipal poverty and prior quotas into exposure. Population weighting emphasizes residents in larger municipalities; it does not turn the pooled coefficient into an effect for every municipality.

Table 8 compares the preferred result with state-by-post effects and an additional old-rule quota adjustment. The vote-share reduced form and IV estimate attenuate under both, even as first stages remain strong. Nonlinear controls and the earlier authorization horizon also yield smaller, less precise electoral estimates. The result survives several outcome and weighting choices in Table 9, but it does not survive every identifying comparison. Table 10 checks pre-reform incumbent vote share; Table 11 examines selection into rerunning. These diagnostics address observable concerns without proving exclusion, monotonicity, or the absence of unobserved electoral trends.

3.5 Heterogeneity and Scope

Municipality Size. The pooled response is concentrated in municipalities of a particular size. The electoral gains are clearest among mayors of larger municipalities. The confidence sets exclude zero in the two upper population tails, while the evidence for the two lower-population samples remains inconclusive. Population weighting makes this distinction relevant to the interpretation of the pooled result: the headline estimate should not be read as the return for a typical mayor or municipality. Table 12 reports estimates below and above the median, below and above the 75th percentile, and above the 90th percentile.[10] These comparisons show where the estimated response appears in the sample. They do not identify the effect of making a municipality larger.

Baseline Coverage and Prior Expansion. Prior coverage may leave less room for the authorization shock to operate. The conditional estimates point to electoral gains below median baseline coverage and remain inconclusive above the median. Below and above median baseline coverage, the estimates are 11.60 ([2.88, 22.95]) and 0.04 ([-1.68, 1.38]), respectively (Table 13). These paired comparisons use conditional inference, which limits what we can conclude about differences between the groups.[11]

[10] Below the median and below the 75th percentile, the IV estimates are -0.03 ([-1.60, 1.26]) and 0.03 ([-1.62, 1.57]). Above the 75th percentile, the reduced form is 2.77 and the IV estimate is 9.67 ([4.22, 18.04]). Above the 90th percentile, they are 5.35 and 17.23 ([8.40, 39.39]).

[11] The baseline-coverage, Frey, region, and party partitions allow instruments, endogenous variables, and the complete control specification to differ across groups. The joint regression retains their covari-

The same table compares municipalities inside and outside the Frey (2019) sample. The outside and inside estimates are 12.84 ([5.64, 26.27]) and 0.06 ([-1.60, 1.65]). The conditional estimates point to electoral gains outside the Frey sample and remain inconclusive inside that sample. The conditional first-stage F statistics are 17.83 and 40.67, respectively. This comparison relies on the enrollment response within each group. Earlier expansion could have reduced the eligible pool still available for enrollment. But coverage in 2008, although measured before our shock, may reflect that earlier expansion as well as other municipal differences. The comparison cannot establish that the program had exhausted the eligible pool or that saturation explains the electoral pattern.

Region and Party. The regional and partisan comparisons locate the pooled return across municipalities and incumbents. Outside and inside the Northeast, the estimates are 2.54 ([-1.30, 8.36]) and 1.23 ([-1.00, 3.27]; Table 14). The regional comparison does not establish a difference in the electoral return under the conventional state-clustered test at the five-percent level. For non-PT and PT incumbents, the estimates are 4.40 ([1.05, 8.30]) and -1.90 ([-84.89, 0.75]). The conditional estimates point to electoral gains for non-PT incumbents and remain inconclusive for PT incumbents. The party-group comparison does not establish a difference in the electoral return under the conventional state-clustered test at the five-percent level. These comparisons describe the reach of the pooled finding; they do not identify how voters divide credit between federal and local politicians.

3.6 Robustness

The end-of-text control comparison adds each baseline control group to the lean specification and removes each group from the complete specification (Tables 33 and 35). The vote-share coefficient moves from 7.39 to 3.08 percentage points across the two headline columns. The first-stage F rises while the reduced form falls. Adding the 2008 benefit value alone to the lean column raises the 2SLS estimate above the lean estimate, whereas removing the mayor block from the complete column also increases it. Thus, the combined adjustment changes the electoral magnitude even though the complete-column Anderson–Rubin set excludes zero. We report all predeclared variants, including attenuated and imprecise estimates, rather than choosing controls for

ance. The reported AR sets hold the other group's coefficient at its estimate; they are conditional diagnostics rather than joint confidence regions robust to weak identification of both coefficients. Differences use conventional state-clustered tests and are not robust to weak instruments.

the most favorable coefficient.

Table 8 holds the complete specification and sample fixed while changing one design feature at a time. State-by-period effects and the old-rule expected quota change substantially attenuate the vote-share 2SLS coefficient, while the first stage strengthens. Nonlinear baseline terms and the 2009-only quota instrument also yield smaller estimates for the same 2008–2012 beneficiary outcome. Their Anderson–Rubin sets include zero. These changes chiefly reflect a smaller reduced form, not a weak first stage. The 2009 comparison uses an earlier authorization horizon and should not replace the cumulative instrument by default. Removing FPM or initial benefit value individually moves the estimate in different directions; controlling for their 2008 levels cannot rule out a direct effect of quota exposure through later fiscal transfers or benefit composition.

Outcome Definitions and Extreme Values. The electoral return could depend on how we measure support or on a few extreme observations. Table 9 addresses both concerns. Winsorizing vote share gives an estimate of 2.79 ([0.66, 5.76]), while using registered voters as the denominator gives 2.77 ([0.49, 5.96]). Winsorizing the authorization shock gives 3.83 ([1.42, 6.85]). The registered-voter measure counts support relative to the electorate, rather than only voters who cast a valid vote for a candidate.

Authorization Horizon. The 2012 quota revision may matter for the expansion observed over the mayoral term. Using only the official 2009 quota relative to the counterfactual allocation gives an estimate of 0.57 ([-4.04, 4.48]). Enrollment still measures the full 2008–2012 change. This comparison asks whether the electoral result depends on the later authorization endpoint; it does not isolate the return to families enrolled during 2008–2009.

Selection into Rerunning. The sample contains mayors who chose to seek reelection. Reweighting for that choice using observed characteristics gives an estimate of 2.74 ([0.39, 5.63]). This adjustment combines inverse-probability and population weights. Its value depends on overlap and on the selection model, which cannot account for unobserved determinants of both rerunning and electoral performance.

Normalization and Discretization. We also examine whether the result depends on the way municipal expansion is measured. Using Census-2000 families as the denom-

inator for enrollment and authorization gives 3.14 ([0.45, 7.00]). This changes the definition of exposure while retaining population weights in estimation.

A separate exercise asks whether authorization moves municipalities across successively higher thresholds of enrollment growth (Table 15). The estimates for crossing the median, 75th-percentile, and 90th-percentile thresholds are 12.81 ([3.21, 25.23]), 11.48 ([3.11, 21.74]), and 15.47 ([3.85, 34.99]). The continuous specification gives 3.08 ([0.74, 6.40]). Because every panel keeps the same authorization instrument, the reduced form remains 1.06 by construction. The threshold estimates refer to different changes in program coverage than the continuous coefficient. Their causal interpretation requires exclusion through the chosen binary treatment, so they do not by themselves recover an unrestricted dose-response relationship.

3.7 What Expanded Access Means for Families

The electoral return accompanies expanded access to a program that can improve families' living conditions. Registration can open a path to assistance, although entering *Cadastro Único* does not guarantee a transfer. For families who receive *Bolsa Família*, the transfer adds income and is linked to health and education conditions. Existing research helps explain why that access could matter to households; it does not estimate the welfare effects of the quota reform studied here.

Health. Rasella et al. (2013) document lower childhood mortality as *Bolsa Família* coverage expands, particularly from poverty-related causes. Their findings suggest that access to the program can have consequences beyond the transfer itself. Our municipal electoral estimates do not measure those health gains or distinguish the contribution of income from that of health services.

Education. Glewwe and Kassouf (2012) find higher school enrollment, lower dropout rates, and greater grade promotion following the introduction of *Bolsa Escola/Bolsa Família*. These outcomes give substance to the value of reaching eligible families with children. They concern schooling under the broader program, however, and do not establish the educational effect of the additional registrations induced by the quota reform.

Income and Poverty. Using household surveys, Ferreira de Souza et al. (2019) show that program transfers reduce measured poverty and inequality when included in

household income. This accounting evidence illustrates the immediate importance of benefit receipt for poor families. Taken together, the literature makes it plausible that expanded access yields material gains alongside the mayor's electoral return. It does not tell us how much of either gain depends on new registration, or whether families reward the mayor for the income, the assistance in obtaining it, or both.

Electoral Returns and Local Implementation. de Janvry et al. (2012) study how mayoral electoral incentives relate to the performance of a decentralized education transfer program. Frey (2019) finds a negative local-incumbency response to transfer expansion in a different set of small, low-HDI municipalities using a health-financing source of variation. His coverage denominator, sample, and treatment margin differ from ours, so the coefficients cannot be compared as effects of the same dose. Frey and Santarrosa (2026) document the political affiliation of household-facing bureaucrats, a related question about implementers rather than the electoral return to an externally induced change in coverage. Our contribution is to estimate how federal authorization translates into local beneficiary expansion and mayoral vote share when municipalities remain responsible for access. We turn next to the municipal institutions through which families encounter the program. The CRAS comparisons ask where that return appears, with the limitations stated below.

4 Possible Mechanisms

The federal government pays the transfer, while municipal staff often help families reach it. We propose that a pre-existing welfare office can make this municipal role visible to families and help translate program expansion into votes for the mayor. The evidence below asks whether enrollment and electoral responses move together across municipalities with different service capacity. Income gains, local economic effects, and changes in the electorate remain possible explanations for the same electoral response.

Families may first encounter the program through a municipal office or an outreach team. Staff can identify a family, explain eligibility, collect information, and direct it to other services. That contact supplies a possible route for local political credit even when benefit values and payments are federal. Evidence from Uruguay shows that transfers can change expressed support for the national government, although its outcome and attribution setting differ from our municipal election (Manacorda et al., 2011).

We use the quota-induced expansion from Section 3 to compare enrollment and electoral gains across municipalities with different welfare infrastructure.[12] Because offices and service practices were not randomly assigned, these comparisons can inform the contact explanation without identifying the effect of providing those services.

4.1 Local Welfare Offices and Electoral Returns

Table 3 shows that quota exposure raises first registrations, with the larger response among families recorded at or below the program's income threshold in 2012. This administrative margin gives the municipal-contact explanation substance. The snapshot does not link each new registration to subsequent transfer receipt or to an individual vote. Table 7 therefore describes a related expansion margin rather than an identified mediation effect.

A *CRAS* gives families a place to seek assistance, meet municipal staff, and obtain help accessing social programs. Of the 2,078 municipalities in our sample, 1,394 already had a recorded center in 2008 and 684 had none recorded. We ask whether mayors gain more votes from program expansion where families already have this point of contact with the administration.

Federal authorization expanded enrollment in municipalities both with and without a welfare center. The first-stage coefficients are 0.34 with a *CRAS* and 0.35 without one. The data do not clearly distinguish the two enrollment responses. Table 16 reports both enrollment and electoral outcomes.[13]

The electoral gains are concentrated in municipalities that already had a welfare center. A one-standard-deviation increase in enrollment induced by the quota shock corresponds to 4.10 percentage points of incumbent vote share with a *CRAS*, compared with -0.98 without one. The authorization estimates follow the same comparison: 1.41 and -0.34 percentage points, respectively.[14]

This contrast directs attention to the municipality's role in families' access to the

[12]The subgroup IV comparisons start from the main sample and retain its scaling, population weights and baseline controls, with further sample restrictions stated below, and allow the instrument, treatment and controls to vary by group. Standard errors are clustered by state. Group-specific Anderson–Rubin sets are conditional diagnostics that hold the other group's coefficient at its estimate; they are not joint confidence regions robust to weak instruments. Conventional difference tests accompany these comparisons. Additional reduced-form diagnostics report state-clustered intervals, wild-cluster tests and the specified multiplicity adjustments.

[13]The test of equal first-stage coefficients gives $p = 0.904$. A failure to distinguish these coefficients does not establish that the enrollment responses are identical.

[14]The conditional AR sets for the vote-share estimates are [1.36, 8.45] with a *CRAS* and [-4.65, 2.49] without one. The conventional test of their difference gives $p = 0.024$.

program. A welfare office may make that role more visible even when families in other municipalities also receive the federal benefit. The comparison is consistent with local contact helping voters associate access to the program with their mayor. Other explanations remain open: municipalities with a center may differ in the families reached by expansion, the services those families receive, or other features of local administration.

Municipalities with and without a recorded office differ in population, poverty, coverage, and other baseline characteristics (Table 17). Table 18 compares their quota responses under population weights and weights that balance seven observed baseline variables and state shares. Enrollment rises in both groups under either weighting scheme. The vote-share contrast is 1.756 percentage points under population weights, with a wild-cluster p-value of 0.054 and a Holm-adjusted p-value of 0.162. After balancing, that contrast falls to 0.266 percentage points, with a wild-cluster p-value of 0.740. The change in weights changes which municipalities matter most and cannot balance unobserved attributes. The initial CRAS pattern motivates the contact explanation, while this sensitivity limits how strongly we can attribute the electoral difference to the office.

4.2 Outreach and Family Contact

Outreach brings municipal staff to families who may not seek assistance at an office. It may make municipal effort visible to people who would otherwise have little contact with the welfare administration. Within municipalities that already had a *CRAS*, we compare those reporting active outreach before the reform with those without recorded outreach. The underlying coding groups missing responses with no recorded outreach, but an audit of the original source finds no missing outreach responses in this estimation sample.

The estimated electoral return is larger where offices report active outreach: 5.14 percentage points, compared with 0.52 without recorded active outreach. The estimates leave considerable uncertainty about the gap between these electoral returns. Table 19 reports the comparison.[15] The first-stage point estimate is smaller in municipalities reporting active outreach.

These offices were already conducting outreach before the reform. The comparison therefore cannot measure an outreach campaign caused by expansion. Their resources, the families they serve, or the visibility of local officials could also account for the dif-

[15]The conditional AR sets are [1.15, 9.70] with outreach and [-2.74, 3.96] without it. The conventional difference test gives $p = 0.099$.

ference in electoral returns.

Reported Contact Practices. Reported outreach provides a more specific prediction: if contact matters, places with active search before the reform might show a larger electoral response to authorization. Table 19 compares these places within the baseline-CRAS sample and reports their enrollment responses as well. The pattern is suggestive, but outreach, the resources that support it, and the populations served are not randomly assigned. Office reports do not measure the encounters experienced by individual voters.

4.3 Family Follow-Up through PAIF

PAIF is the family-support program delivered through CRAS, known before the reform as the *Programa de Atenção Integral à Família* (Comprehensive Family Support Program). It assists families facing social vulnerability and helps them access benefits and public services.[16] Follow-up may keep families in contact with municipal staff after they first seek assistance, giving them repeated opportunities to observe the municipality's role. We use the pre-reform record of *Bolsa Família* families served by welfare offices as a proxy for PAIF activity, although this count does not measure how often each family meets staff.

The estimated electoral return is larger where offices record families served through PAIF: 4.85 percentage points, compared with 1.68 without a positive recorded family count. The estimates leave considerable uncertainty about the gap between these electoral returns. Table 20 reports this comparison.[17] The first-stage point estimate is smaller in municipalities reporting PAIF follow-up.

We also distinguish municipalities by the recorded number of beneficiary families served, summed across their welfare offices. This comparison uses the 962 municipalities with a positive recorded count, divided at their pre-reform median; it therefore concerns a narrower population than the preceding comparison. The estimated electoral return is larger in the higher-volume group: 12.90 percentage points, compared with -1.98 in the lower-volume group. The evidence supports a difference in electoral

[16]The Ministry's institutional history and description of PAIF dates its establishment to 2004. In 2009 it became the *Serviço de Proteção e Atendimento Integral à Família* (Comprehensive Family Protection and Support Service), retaining the acronym. Its activities include family support and referrals to the registry and other public services. The English names here describe the Portuguese titles.

[17]The conditional AR sets are [0.48, 10.45] with recorded PAIF follow-up and [-2.13, 8.03] without it. The conventional difference test gives $p = 0.355$. The conditional first-stage F statistics are 41.70 and 9.53, respectively.

returns between these groups. Table 21 presents the estimates.[18] The recorded family count does not measure the quality of assistance or the amount of contact received by each family.

4.4 Institutional Exposure and Network Expansion

We harmonize the official CRAS censuses from 2007 through 2012, retaining both the office counts observed in each wave and the reported opening dates.[19]

An office that has operated longer may create repeated opportunities for contact. Table 22 compares offices opened before 2007, offices opened in 2007–2008, and municipalities without a recorded baseline office. Table 23 then describes later opening cohorts. The estimated electoral response tends to be larger around earlier offices, but adjusted tests do not establish a steady increase with office age. Reported opening dates can refer to predecessor facilities, and post-reform openings may respond to municipal decisions associated with the shock. Office duration therefore refines the institutional hypothesis without identifying an exposure effect.

Network Expansion and Identification. An office can first appear in a census wave without having just opened; nonresponse cannot be coded as closure. The pre-reform measure therefore uses offices recorded before the reform, while later cohorts are descriptive. Authorization could itself stimulate new local services. If those services affect votes independently of measured beneficiary receipt, they would be another route from the quota shock to elections. The quota-to-enrollment IV would then summarize a broader local response unless that route is ruled out.

4.5 Alternative Explanations

Material Benefits and Political Credit. Additional income could change how families evaluate government, independently of the office that helped them register. In

[18] The conditional AR sets are [3.18, 26.56] in the higher-volume group and [-7.77, 3.60] in the lower-volume group. The conventional difference test gives $p = 0.044$. The lower group includes observations at the median. Both PAIF comparisons use a family-count item that lacks its complete original questionnaire label in our extract. Its interpretation as PAIF follow-up remains provisional. The binary comparison codes missing counts together with zeros. The count is a proxy for service activity, not a verified measure of repeated contact.

[19] The Ministry of Social Development provides the annual data, questionnaires and manuals through its Censo SUAS archive. Baseline duration uses dates reported in 2008, rather than dates recalled after the reform. Source questionnaires and manuals distinguish registry referral from registration and identify the list-use items used in the contact comparisons.

Uruguay, Manacorda et al. (2011) find persistent government support after transfers end and discuss both beliefs about government and reciprocity as explanations. Our municipal election data do not reveal these beliefs or whom families credit for assistance. We also cannot separate broader local economic effects on voters who do not receive transfers. Electoral gains could therefore arise without an encounter with municipal staff.

Other Registry-Linked Programs and Fiscal Channels. During this period, registration could also affect access to other federal programs, including the social electricity tariff.[20] If new registration generated a valued service independently of *Bolsa Família* receipt, the quota shock could affect votes through that route. Quota exposure could also be correlated with fiscal transfers or new local services; Table 24 examines measured alternative channels. The design does not quantify their separate contribution. The beneficiary IV therefore requires a stronger exclusion interpretation than the direct authorization-to-votes reduced form.

Participation and Electoral Composition. A mayor's vote share can rise when voters switch candidates, when turnout changes, or when the registered electorate changes. Table 25 places incumbent votes, other candidate votes, invalid ballots, and abstention on a common electoral denominator; Table 26 applies the IV comparison. Incumbent votes rise relative to other candidates, and the registered electorate responds to quota exposure. Aggregate totals cannot identify which individuals moved between categories. The pattern is consistent with a change in relative support and with participation or electorate composition. The conflicting Progresa evidence also cautions against inferring individual mobilization from aggregate comparisons (De La O, 2013; Imai et al., 2020).

Complementary Services and Political Relationships. Offices that register families also provide assistance that families may value in its own right. The PAIF comparisons cannot separate appreciation of those services from credit for access to the federal transfer. Differences in who becomes enrolled could also explain why electoral gains vary across municipalities, even if contact has no independent effect. The sensitivity to municipal composition makes this possibility relevant. Political relationships may also

[20] Law No. 12,212/2010 linked eligibility for the social electricity tariff to registration in *Cadastro Único* under its income criteria. This establishes a potential route, not a measured response of tariff receipt to the quota shock.

change as families become less economically vulnerable: Frey (2019) interprets a negative local-incumbency response to transfer expansion through reduced opportunities for clientelistic exchange. Our positive estimate concerns a different source of expansion and population, and allows for offsetting changes in such relationships. We do not observe exchanges of benefits for votes.

4.6 Institutional Interpretation

The institutional link is between federal authorization and local access. The federal government finances the benefit, but municipal staff can help eligible families become visible to the program by identifying them, explaining its requirements, and assisting with registration. The increase in first registrations among families recorded within the income eligibility range is consistent with this role. It shows that expansion reaches beyond the registry's existing stock. It does not establish that all additional beneficiaries were newly registered: families already in the registry could also obtain benefits when quotas expanded.

This local role offers a plausible connection between household gains and electoral support. A family may value both the transfer and the assistance that makes access possible. Where an office or outreach worker provides that assistance, the municipality's contribution may be visible even though the payment comes from the federal government. The CRAS and outreach comparisons motivate this interpretation, while the composition sensitivity and the less consistent evidence on contact practices and operating duration limit what we can attribute to those institutions. The comparisons support studying local assistance as a possible mechanism; they do not isolate the political return to a particular encounter or service.

The household evidence discussed in Section 3.7 explains why facilitating access could have value for families as well as for officeholders. Electoral rewards and improved access to social protection can coexist. Our results are consistent with that alignment, but do not show how much municipal effort was motivated by the prospect of reelection. Nor do they identify the share of transfers or votes that would disappear if new registration had not occurred. The conditional registration scenarios therefore describe the consequences of assumed links between registration, receipt, and votes; they do not supply the missing causal decomposition. The substantive finding is that federal expansion has a local political return in a setting where municipal assistance can help families obtain a valuable benefit.

5 Conclusion

This paper studies the electoral consequences of expanding a federal cash-transfer program whose implementation depends on municipal governments. We use changes in federal enrollment quotas to examine how local expansion of *Bolsa Família* affects support for mayors seeking reelection. Greater authorization brought more families into the program and increased incumbent vote share. Our preferred population-weighted estimate implies an electoral gain of 3.08 percentage points for a one-standard-deviation increase in enrollment induced by the quota reform.

The findings highlight two distinct consequences of the expansion. Federal authorization increased the number of beneficiary families and was associated with additional low-income families entering the registry; the estimated electoral return accrued to mayors who sought reelection. Registry entry can improve access to assistance, while the broader literature documents benefits of program receipt for families. Our data do not link a newly registered family to its subsequent payment or vote, and they do not measure the welfare effect of this particular expansion. The evidence on local welfare offices is consistent with contact between families and municipal staff contributing to electoral support, but the office comparisons do not identify the effect of installing a *CRAS*.

The broader implication is that the electoral consequences of redistribution depend on how programs reach their beneficiaries. Centralized rules and payments leave an important role for local governments in making benefits accessible. The electoral estimate is sensitive to state-specific trends and allocation-rule controls, and the exclusion restriction cannot be verified from these data. Within those limits, the pattern shows how a federally financed expansion can enlarge access for vulnerable families while giving local incumbents an electoral return.

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Figures and Tables

Table 1: *Bolsa Família* Eligibility Thresholds and Benefit Values

	2004 Jan.	2006 Apr.	2007 Aug.	2008 Mar.	2008 July	2009 Apr.	2009 July	2009 Sept.	2011 Apr.	2011 Sept.	2011 Nov.	2011 Dec.	2012 June	2012 Dec.	2013 Mar.	2014 May	2014 June	2016 June	2016 July	2018 July
Panel A: Income Eligibility Thresholds (BRL per capita)																				
Extreme Poverty Line	50	60	60	60	60	69	70	70	70	70	70	70	70	70	70	77	77	85	85	89
Poverty Line	100	120	120	120	120	137	140	140	140	140	140	140	140	140	140	154	154	170	170	178
Panel B: Benefit Values (BRL)																				
Basic Benefit (per family)	50	50	58	58	62	62	62	68	70	70	70	70	70	70	70	70	77	77	85	89
Variable Benefit (per child aged 0–15)	15	15	18	18	20	20	20	22	32	32	32	32	32	32	32	32	35	35	39	41
Youth Benefit (per child aged 16–17)	–	–	–	30	30	30	30	33	38	38	38	38	38	38	38	38	42	42	46	48
Nursing Benefit (per child aged 0–6 months)	–	–	–	–	–	–	–	–	–	–	32	32	32	32	32	32	35	35	39	41
Pregnancy Benefit (per pregnancy)	–	–	–	–	–	–	–	–	–	–	–	32	32	32	32	32	35	35	39	41
Panel C: Family Benefit Limits (per family)																				
Maximum Benefit (BRL)	95	95	112	172	182	182	182	200	242	306	306	306	306	306	306	306	336	336	372	390
Variable Benefits	3	3	3	3	3	3	3	3	3	5	5	5	5	5	5	5	5	5	5	5
Youth Benefits	–	–	–	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2

Notes: Columns cover 2004–2018 and report effective months: legal dates for income thresholds and first payment months for benefit changes. Values are nominal BRL; unchanged rules carry forward through December 2018. A dash indicates a component not yet implemented. The basic benefit applies only to extremely poor families. Maximum Benefit is the largest monthly sum of the basic benefit, the capped variable benefits and the separately capped youth benefits for an eligible family with the requisite composition; it is not a universal ceiling on actual transfers. Variable Benefits is the shared family limit for children aged 0–15, pregnancies and nursing children; Youth Benefits has a separate limit of two per family. Pregnancy and nursing benefits pay nine and six monthly installments per eligible episode, respectively. The family-specific extreme-poverty supplement introduced in 2012 and legacy-program compensation are excluded from this display and from Maximum Benefit; subnational supplements are also excluded.

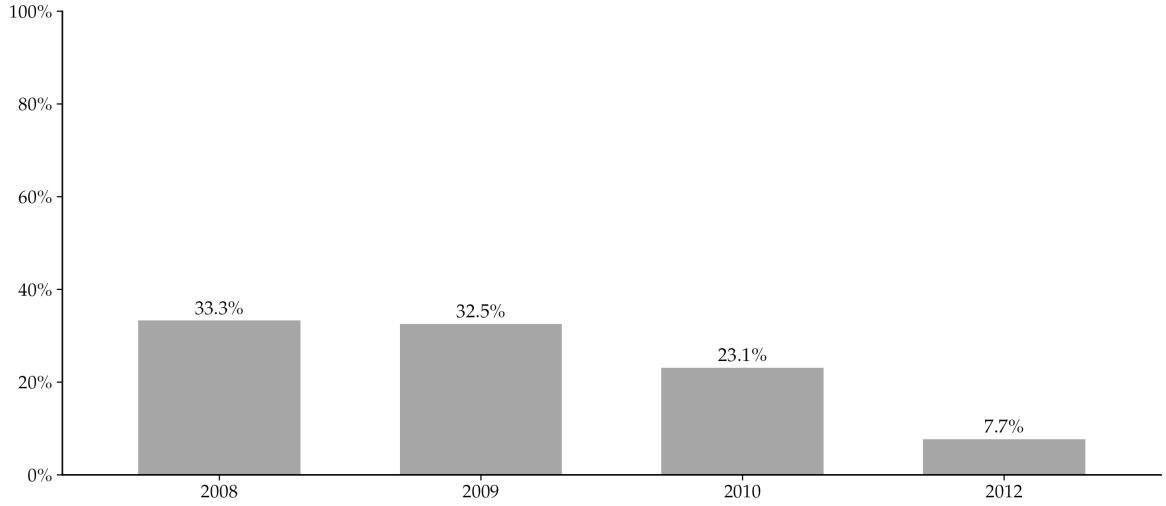
Sources: Jan. 2004 baseline: Law 10,836. Apr. 2006 thresholds: Decree 5,749; Aug. 2007 amounts: Decree 6,157. Mar. 2008 youth benefit: Provisional Measure 411 and MDS Informe 158; July 2008 amounts: Decree 6,491. Apr. 2009 thresholds: Decree 6,824; July thresholds/Sept. amounts: Decree 6,917. Apr. 2011 amounts: Decree 7,447; Sept. family limit: Decree 7,494 and MDS Informe 285; Nov./Dec. nursing/pregnancy implementation: MDS Informe 293. 2012–13 extreme-poverty supplement: Decree 7,758, Decree 7,852 and Law 12,817 (art. 2). May thresholds/June amounts, 2014: Decree 8,232; June thresholds/July amounts, 2016: Decree 8,794; July 2018 thresholds and amounts: Decree 9,396.

Table 2: Descriptive Statistics

All Municipalities	N	Min	Mean	Median	Max	SD	**Regression Sample**	N	Min	Mean	Median	Max	SD
Panel A: Census Information							**Panel A:** Census Information						
Population (2000)	4,976	795.00	31,515.59	10,703.50	10,277,518.00	190,735.22	Population (2000)	2,078	795.00	27,991.14	10,556.50	5,773,130.00	146,177.39
Urban Population (2000) (%)	4,976	0.00	59.29	59.92	100.00	23.32	Urban Population (2000) (%)	2,078	0.00	58.80	59.35	100.00	23.32
Poor Population (2000) (%)	4,976	5.17	64.02	66.23	99.00	20.76	Poor Population (2000) (%)	2,078	7.66	64.09	66.80	99.00	20.99
Income per capita (2000) (R$)	4,976	62.65	338.27	306.00	1,759.76	193.29	Income per capita (2000) (R$)	2,078	69.20	337.40	300.13	1,759.76	195.41
Gini (2000) (%)	4,976	30.00	54.78	55.00	82.00	6.80	Gini (2000) (%)	2,078	30.00	54.84	55.00	80.00	6.73
Formal Workers (2000) (%)	4,976	1.92	36.15	34.44	86.38	18.16	Formal Workers (2000) (%)	2,078	2.89	35.80	33.98	86.38	18.31
Panel B: *Bolsa Família* Program Information							**Panel B:** *Bolsa Família* Program Information						
Permitted *Bolsa Família* Families (2008)	4,976	24.00	2,078.06	917.50	285,060.00	6,980.68	Permitted *Bolsa Família* Families (2008)	2,078	24.00	1,898.41	906.00	164,913.00	5,077.26
Permitted *Bolsa Família* Families (2008) (%)	4,976	0.77	8.81	8.29	67.43	4.66	Permitted *Bolsa Família* Families (2008) (%)	2,078	0.77	8.89	8.39	33.00	4.71
Permitted *Bolsa Família* Families (2008) per 10,000 Residents	4,976	77.08	880.65	829.13	6,742.82	466.41	Permitted *Bolsa Família* Families (2008) per 10,000 Residents	2,078	77.08	888.88	839.23	3,299.89	470.55
Δ Permitted *Bolsa Família* Families per 10,000 Residents	4,976	-2,841.24	116.55	91.84	2,865.11	274.55	Δ Permitted *Bolsa Família* Families per 10,000 Residents	2,078	-1,000.62	120.42	95.75	2,865.11	277.26
Realized *Bolsa Família* Families (2008)	4,976	2.42	2,045.08	914.79	204,410.17	5,990.93	Realized *Bolsa Família* Families (2008)	2,078	5.50	1,871.27	908.38	133,322.75	4,543.18
Realized *Bolsa Família* Families (2008) (%)	4,976	0.08	8.76	8.13	61.49	4.91	Realized *Bolsa Família* Families (2008) (%)	2,078	0.26	8.80	8.31	29.64	4.90
Realized *Bolsa Família* Families (2008) per 10,000 Residents	4,976	8.48	876.40	813.07	6,149.19	491.21	Realized *Bolsa Família* Families (2008) per 10,000 Residents	2,078	25.81	879.87	830.99	2,963.95	490.46
Δ Realized *Bolsa Família* Families per 10,000 Residents	4,976	-1,234.13	180.99	139.04	2,979.12	231.70	Δ Realized *Bolsa Família* Families per 10,000 Residents	2,078	-398.59	203.50	163.15	2,979.12	238.38
Panel C: Election Information							**Panel C:** Election Information						
Candidates (2008)	4,976	2.00	2.73	2.00	12.00	1.02	Candidates (2008)	2,078	2.00	2.80	3.00	12.00	1.07
Candidates (2012)	4,976	2.00	2.77	2.00	12.00	1.08	Candidates (2012)	2,078	2.00	2.70	2.00	9.00	1.00
Seeking Reelection (2012) (%)	4,976	0.00	43.75	0.00	100.00	49.61	Seeking Reelection (2012) (%)	2,078	100.00	100.00	100.00	100.00	0.00
Winner's Vote Share (2008) (%)	4,976	24.06	54.97	53.73	99.98	10.46	Winner's Vote Share (2008) (%)	2,078	24.06	52.92	52.68	99.16	9.43
Incumbent Vote Share (2012) (%)	2,177	0.00	48.28	49.24	100.00	16.41	Incumbent Vote Share (2012) (%)	2,078	0.18	49.99	49.65	100.00	14.03
Δ Incumbent Vote Share (pp)	2,177	-79.46	-4.60	-3.60	55.47	17.29	Δ Incumbent Vote Share (pp)	2,078	-71.82	-2.94	-3.03	55.47	15.13
Panel D: Candidate Information							**Panel D:** Candidate Information						
Female Mayor (2008) (%)	4,970	0.00	9.22	0.00	100.00	28.93	Female Mayor (2008) (%)	2,078	0.00	10.15	0.00	100.00	30.21
High School or More (2008) (%)	4,970	0.00	78.85	100.00	100.00	40.84	High School or More (2008) (%)	2,078	0.00	81.14	100.00	100.00	39.13
College Degree (2008) (%)	4,970	0.00	45.11	0.00	100.00	49.77	College Degree (2008) (%)	2,078	0.00	46.39	0.00	100.00	49.88
Mayor Age (2008)	4,970	21.00	48.16	48.00	90.00	9.52	Mayor Age (2008)	2,078	21.00	46.51	46.00	81.00	9.08
Married (2008) (%)	4,970	0.00	78.13	100.00	100.00	41.34	Married (2008) (%)	2,078	0.00	77.05	100.00	100.00	42.06
Public-Sector Occupation (2008) (%)	4,970	0.00	34.21	0.00	100.00	47.44	Public-Sector Occupation (2008) (%)	2,078	0.00	15.74	0.00	100.00	36.42
Agricultural Occupation (2008) (%)	4,970	0.00	12.03	0.00	100.00	32.54	Agricultural Occupation (2008) (%)	2,078	0.00	14.49	0.00	100.00	35.20
Business Occupation (2008) (%)	4,970	0.00	20.24	0.00	100.00	40.18	Business Occupation (2008) (%)	2,078	0.00	26.32	0.00	100.00	44.05

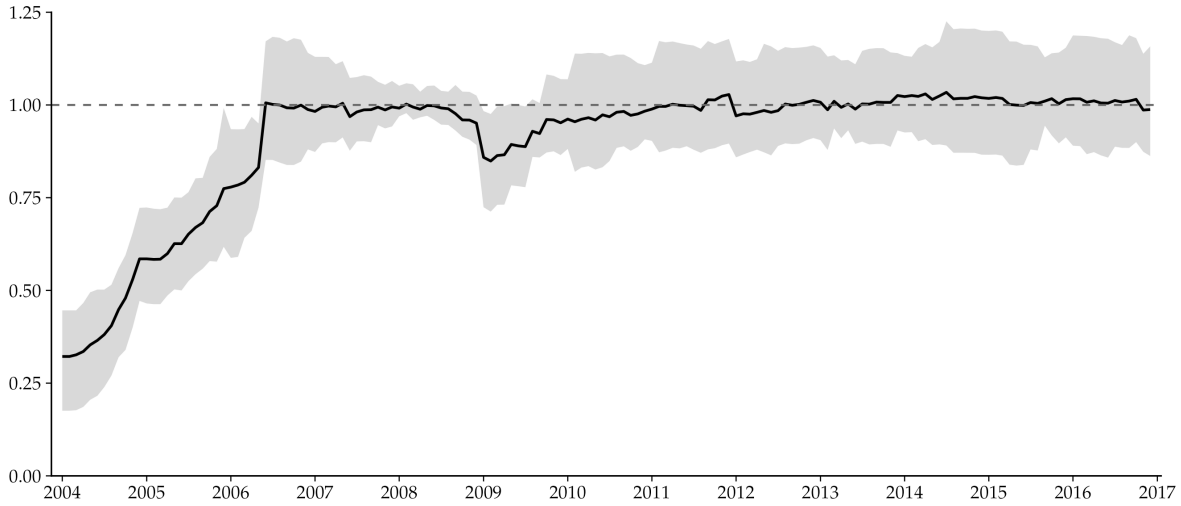
Notes: Each observation is one municipality. All Municipalities contains municipalities with a balanced 2008–2012 election panel that satisfy the common election-quality restrictions; N is the number with nonmissing values in each row. Regression Sample is the exact locked regression sample used in the headline estimates (2,078 municipalities). Permitted families in 2008 are the pre-reform authorized quota (EST 2006); realized families are 12-month averages. Percentage and per-10,000 measures use fixed 2006 population. Every Δ is the corresponding 2012 value minus its 2008 value. Candidate counts describe first-round mayoral candidates. Seeking reelection indicates that the sitting mayor contested the 2012 election. Vote share tracks the 2008 winner: the 2008 value is that winner's baseline share, and the 2012 value and its change are observed only when the same mayor reran. Mayor characteristics describe the 2008 winner and match the headline controls; age squared is omitted because it is a deterministic transformation of age.

Figure 1: Share of Municipalities with Registrations Below Authorized Quotas



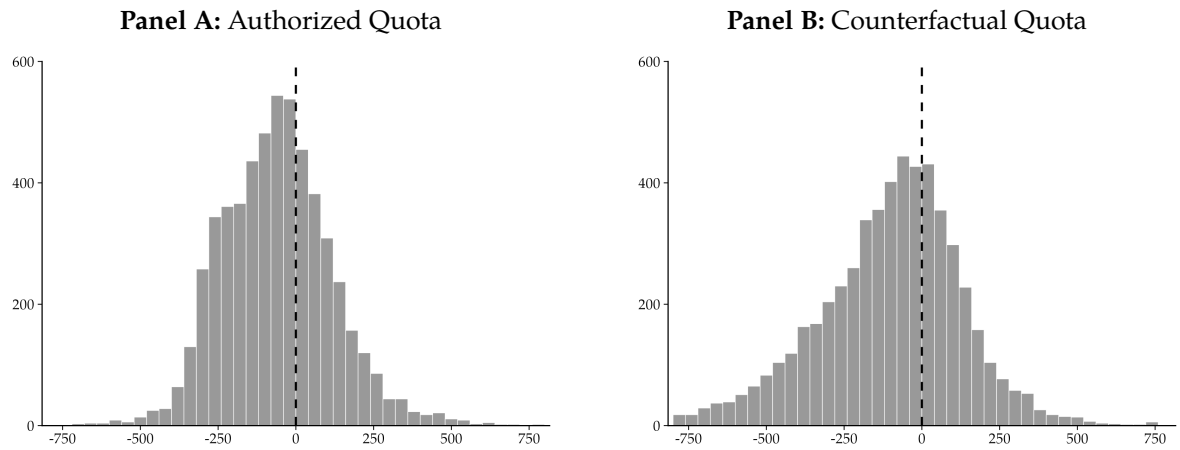
Notes: Each bar is the unweighted share of 5,564 municipalities where the cumulative number of families in the 2012 *Cadastro Único* snapshot with reported per-capita income at or below R$140 and a first-registration date no later than that year is strictly below the municipality's December authorized *Bolsa Família* quota. The 2008 count includes registrations dated 2008 or earlier. The source aggregate does not separate 2006 or 2007, so the four displayed years are 2008, 2009, 2010, and 2012. Income is measured in 2012, not necessarily at first registration; families who exited the register before the 2012 snapshot are absent. These bars do not measure benefit take-up.

Figure 2: Realized Enrollment as a Share of Authorized Quota



Notes: For each municipality-month, we compute the ratio of realized *Bolsa Família* beneficiaries to the authorized quota. The solid line plots the quota-weighted cross-municipal mean of these municipality-level ratios. The shaded band spans the weighted 25th to 75th percentile. The dashed horizontal line marks full take-up (ratio = 1). Before 2007, municipalities operated well below their authorized ceilings, and the rapid convergence toward 1 reflects the administrative build-up of the program in its first years. After this initial convergence, realized enrollment generally tracked authorized capacity closely. The 2009 reform temporarily broke that alignment: expanded quotas opened a gap between authorized slots and realized beneficiaries, creating the local implementation margin through which municipalities had to identify, register, and enroll additional eligible families.

Figure 3: Distribution of Slack: Authorized Quota vs. Counterfactual Quota



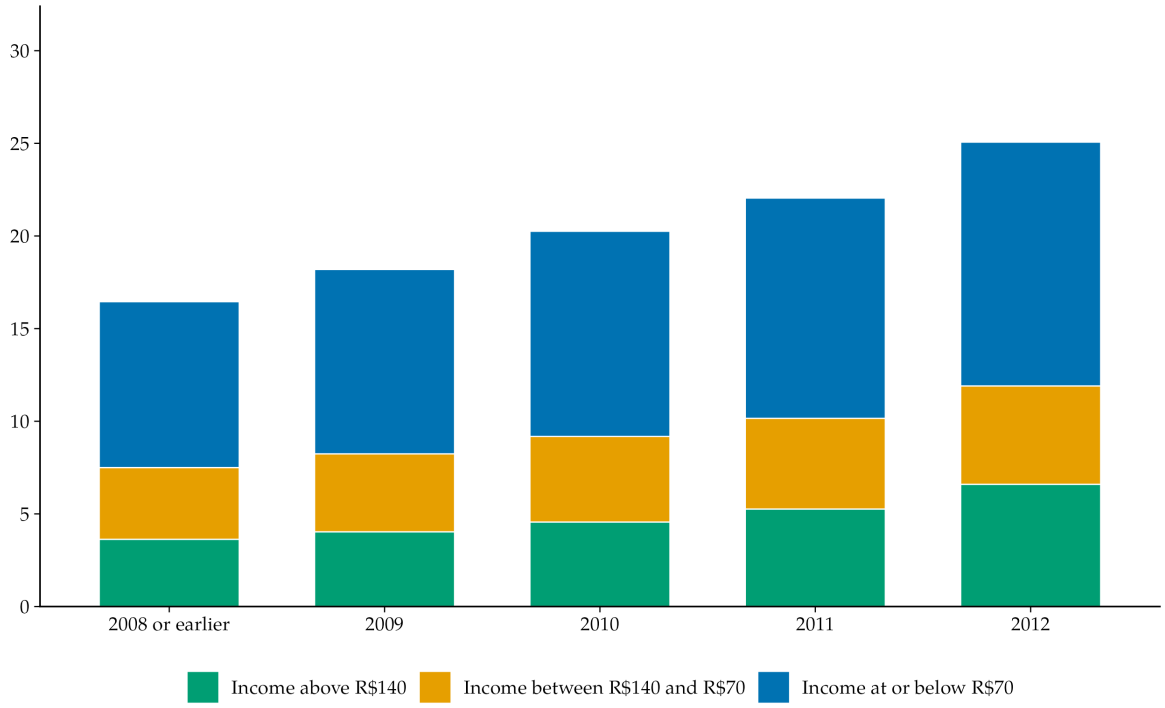
Notes: Each panel shows slack in December 2012 per 10,000 residents using fixed 2006 population. Slack is quota minus the 12-month average stock of beneficiary families in 2012 (not realized enrollment minus quota). Panel A uses the official authorized 2012 quota; Panel B uses the counterfactual 2009 quota implied by the previous allocation rule. Positive values indicate authorized room not translated into realized enrollment, while negative values indicate realized enrollment above the relevant quota. The dashed vertical line marks zero slack. Both panels retain the common display window from -800 to 800 beneficiary families per 10,000 residents to limit the influence of extreme observations.

Table 3: Registration Expansion in *Cadastro Único*

	Δ Registrations in *Cadastro Único*	
	(1)	(2)
Panel A: All registered families		
ΔQuota_m (π_1)	34.464***	48.471***
	(9.096)	(9.465)
Kleibergen–Paap F	14.36	26.22
Panel B: Income above R$140		
ΔQuota_m (π_1)	4.020	4.870
	(3.467)	(3.649)
Kleibergen–Paap F	1.34	1.78
Panel C: Income at or below R$140		
ΔQuota_m (π_1)	30.444***	43.601***
	(7.570)	(8.116)
Kleibergen–Paap F	16.17	28.86
Panel D: Income at or below R$70		
ΔQuota_m (π_1)	24.910***	34.323***
	(6.285)	(7.119)
Kleibergen–Paap F	15.71	23.25
Bolsa Família Characteristics	✓	✓
Party Characteristics	✓	✓
Municipality Characteristics		✓
Mayor Characteristics		✓
Number of Mayors	2,078	2,078
Observations	4,156	4,156

Notes: All columns retain the locked 2008–2012 sample, quota-shock instrument, municipality and year fixed effects, 2006-population weights and state-clustered CRV1 standard errors. Controls are interacted with post. Registration expansion counts families first registered in 2009–2012 and still observed in the 2012 *Cadastro Único* snapshot. Income (monthly per capita) and municipality are measured in 2012. These are surviving entry cohorts, not historical net stocks or verified *Bolsa Família* recipients. The income definitions overlap: the at-or-below-R$140 group combines the two lower-income bands. Each coefficient reports the change in family registrations per 10,000 residents (fixed 2006 population) induced by one weighted SD of the quota shock. Natural-rate first stages add across disjoint income groups. Parentheses report standard errors. Stars use null-imposed state wild-cluster bootstrap p-values from 9,999 draws: $*\ p < 0.10$, $**\ p < 0.05$, $***\ p < 0.01$.

Figure 4: Family Registrations in the 2012 *Cadastro Único* Snapshot by Reported Income



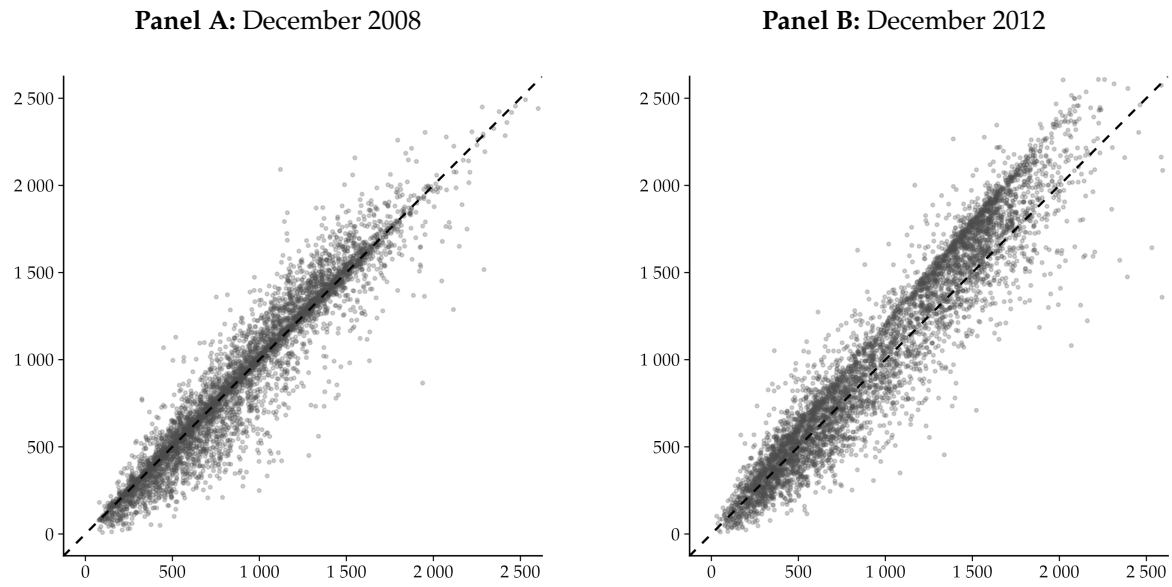
Notes: The vertical scale is in millions of families. Counts sum municipality-level family records in the 2012 snapshot. "2008 or earlier" pools all valid registration dates through December 2008. Municipality and nominal monthly per-capita family income are recorded in 2012; income is not measured at registration. Registration dates outside 1900–2012 are excluded. Each bar cumulates families through the indicated registration year, conditional on their presence in the 2012 snapshot. Stacked segments are mutually exclusive 2012 income categories: at most R$70, above R$70 through R$140, above R$140, and missing or invalid income. These are retrospective cohorts, not historical registry stocks. The available aggregates pool years through 2008 and contain no registrations after 2012; they cannot support separate 2004–2008 bars or a series through 2016.

Table 4: First-Stage Estimates

	ΔBeneficiaries$_m$	
	(1)	(2)
ΔQuota$_m$ (π_1)	0.252***	0.343***
	(0.051)	(0.053)
Kleibergen–Paap F	24.26	41.39
Bolsa Família Characteristics	✓	✓
Party Characteristics	✓	✓
Municipality Characteristics		✓
Mayor Characteristics		✓
Number of Mayors	2,078	2,078
Observations	4,156	4,156

Notes: All columns absorb municipality and year effects. All columns use 2006-population weights and state-clustered standard errors. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. Column (1) includes 2008 incumbent party, 2006 PT presidential vote share, baseline 2008 program intensity and per-family benefit value, and log 2006 quota. Column (2) adds municipality and mayor characteristics, including baseline 2008 FPM transfers in the municipality group. Predetermined controls are interacted with post. Standard errors are in parentheses. Stars denote two-sided null-imposed state wild-cluster bootstrap p-values from 9,999 Rademacher draws: $*$ $p < 0.10$, $**$ $p < 0.05$, $***$ $p < 0.01$.

Figure 5: Authorized Quota vs. Realized Enrollment



Notes: Each dot is a municipality. The horizontal axis shows the federally authorized quota and the vertical axis shows the 12-month average stock of families enrolled in *Bolsa Família*, both per 10,000 residents using fixed 2006 population. The dashed line marks the 45-degree locus of full take-up (realized = authorized). The panels show December 2008, before the mayoral term, and December 2012, at its endpoint. Both axes share a display limit rounded up from the 99th percentile to the next 500-family tick; observations above this limit are excluded from the frame but not from any analysis.

Table 5: Reduced-Form Estimates

	Δ Incumbent's Vote Share		Δ Reelection Probability		Δ Turnout	
	(1)	(2)	(3)	(4)	(5)	(6)
ΔQuota_m (ρ_1)	1.863**	1.056**	3.103	1.409	-0.398***	-0.328*
	(0.571)	(0.381)	(2.056)	(1.443)	(0.195)	(0.196)
Bolsa Família Characteristics	✓	✓	✓	✓	✓	✓
Party Characteristics	✓	✓	✓	✓	✓	✓
Municipality Characteristics		✓		✓		✓
Mayor Characteristics		✓		✓		✓
Number of Mayors	2,078	2,078	2,078	2,078	2,078	2,078
Observations	4,156	4,156	4,156	4,156	4,156	4,156

Notes: All columns absorb municipality and year effects. All columns use 2006-population weights and state-clustered standard errors. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. Column (1) includes 2008 incumbent party, 2006 PT presidential vote share, baseline 2008 program intensity and per-family benefit value, and log 2006 quota. Column (2) adds municipality and mayor characteristics, including baseline 2008 FPM transfers in the municipality group. Predetermined controls are interacted with post. Standard errors are in parentheses. Stars denote two-sided null-imposed state wild-cluster bootstrap p-values from 9,999 Rademacher draws: $*\ p < 0.10$, $**\ p < 0.05$, $***\ p < 0.01$.

Table 6: Two-Stage Least Squares Estimates

	Δ Incumbent's Vote Share		Δ Reelection Probability		Δ Turnout	
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	7.395†	3.084†	12.315	4.114	-1.578	-0.956
	(2.932)	(1.282)	(8.159)	(4.233)	(0.896)	(0.610)
AR 95% CI	[2.442, 16.315]	[0.742, 6.400]	[-4.933, 32.154]	[-4.751, 13.665]	[-4.167, 0.015]	[-2.458, 0.214]
Bolsa Família Characteristics	✓	✓	✓	✓	✓	✓
Party Characteristics	✓	✓	✓	✓	✓	✓
Municipality Characteristics		✓		✓		✓
Mayor Characteristics		✓		✓		✓
Number of Mayors	2,078	2,078	2,078	2,078	2,078	2,078
Observations	4,156	4,156	4,156	4,156	4,156	4,156

Notes: All columns absorb municipality and year effects. All columns use 2006-population weights and state-clustered standard errors. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. Column (1) includes 2008 incumbent party, 2006 PT presidential vote share, baseline 2008 program intensity and per-family benefit value, and log 2006 quota. Column (2) adds municipality and mayor characteristics, including baseline 2008 FPM transfers in the municipality group. Predetermined controls are interacted with post. Standard errors are in parentheses. Reelection probability is conditional on the incumbent rerunning in 2012. Anderson–Rubin 95% confidence sets use quadratic state-clustered inversion with a t_{G-1} critical value and are robust to arbitrary instrument weakness. † attached to a 2SLS coefficient means that its unrounded Anderson–Rubin 95% confidence set excludes zero.

Table 7: Registration Expansion and Electoral Outcomes

	Δ Incumbent's Vote Share		Δ Reelection Probability		Δ Turnout	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Reduced Form						
ΔQuota_m (ρ_1)	1.863**	1.056**	3.103	1.409	-0.398***	-0.328*
	(0.571)	(0.381)	(2.056)	(1.443)	(0.195)	(0.196)
Panel B: All registered families						
$\Delta\widehat{\text{Registrations}}_m$ (β_1)	12.909†	5.204†	21.499	6.944	-2.756	-1.614
	(5.601)	(2.260)	(15.303)	(7.180)	(1.654)	(1.052)
AR 95% CI	[4.131, 34.422]	[1.236, 11.720]	[-8.211, 68.954]	[-8.188, 24.183]	[-8.678, 0.025]	[-4.452, 0.361]
Panel C: Income above R$140						
$\Delta\widehat{\text{Registrations}}_m$ (β_1)	49.826†	23.322†	82.979	31.114	-10.636	-7.230
	(45.832)	(17.892)	(95.490)	(36.378)	(10.060)	(6.302)
AR 95% CI	$(-\infty, \infty)$	$(-\infty, \infty)$	$(-\infty, \infty)$	$(-\infty, \infty)$	$(-\infty, \infty)$	$(-\infty, \infty)$
Panel D: Income at or below R$140						
$\Delta\widehat{\text{Registrations}}_m$ (β_1)	12.353†	4.891†	20.572	6.525	-2.637	-1.516
	(5.278)	(2.181)	(14.251)	(6.815)	(1.606)	(1.009)
AR 95% CI	[3.957, 31.427]	[1.125, 11.174]	[-8.072, 61.340]	[-7.457, 23.044]	[-8.210, 0.023]	[-4.264, 0.327]
Panel E: Income at or below R$70						
$\Delta\widehat{\text{Registrations}}_m$ (β_1)	14.603†	6.010†	24.320	8.018	-3.117	-1.863
	(6.251)	(2.767)	(16.862)	(8.397)	(1.929)	(1.268)
AR 95% CI	[4.685, 37.493]	[1.361, 14.553]	[-9.571, 73.065]	[-9.282, 29.166]	[-9.991, 0.027]	[-5.556, 0.395]
Bolsa Família Characteristics	✓	✓	✓	✓	✓	✓
Party Characteristics	✓	✓	✓	✓	✓	✓
Municipality Characteristics		✓		✓		✓
Mayor Characteristics		✓		✓		✓
Number of Mayors	2,078	2,078	2,078	2,078	2,078	2,078
Observations	4,156	4,156	4,156	4,156	4,156	4,156

Notes: All columns use the fixed 2,078-mayor sample, absorb municipality and year effects, use 2006-population weights, and cluster standard errors by state. The quota instrument and registration changes are rates per 10,000 residents using fixed 2006 population. The instrument is reported per one analysis-weighted SD; each endogenous registration measure uses its own fixed analysis-weighted SD; the measure-specific first-stage estimates appear in Appendix Table 3. Columns (1), (3), and (5) include 2008 incumbent party, 2006 PT presidential vote share, baseline 2008 program intensity and per-family benefit value, and log 2006 quota. Columns (2), (4), and (6) add municipality and mayor characteristics, including baseline 2008 FPM transfers in the municipality group. Reelection probability is conditional on the incumbent rerunning in 2012. Predetermined controls are interacted with post. Standard errors are in parentheses. Registration expansion counts families first registered in 2009–2012 and still observed in the 2012 *Cadastro Único* snapshot; income and municipality are measured in 2012. These are surviving entry cohorts, not historical net stocks or verified *Bolsa Família* recipients. Income definitions overlap: the at-or-below-R$140 group includes the at-or-below-R$70 group. The reduced form is common across panels. These IVs do not separate registration from subsequent transfers. Anderson–Rubin 95% confidence sets use quadratic state-clustered inversion with a t_{G-1} critical value and are robust to arbitrary instrument weakness. Disconnected two-tail sets use the conservative full-real-line envelope; exact components remain in the result files. Reduced-form stars use null-imposed state wild-cluster bootstrap p-values: $*\ p < 0.10$, $**\ p < 0.05$, $***\ p < 0.01$. † on a 2SLS estimate denotes exclusion of zero by its unrounded AR 95% set.

Table 8: Identification Sensitivities

	Δ Incumbent's Vote Share					
	Main	State Trends	GNS (2026)	Nonlinear	w/o FPM Transfers	w/o Benefit Value
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: First Stage						
ΔQuota_m (π_1)	0.343***	0.510***	0.556***	0.388***	0.333***	0.401***
	(0.053)	(0.041)	(0.035)	(0.050)	(0.053)	(0.063)
Kleibergen–Paap F	41.4	153.1	251.8	59.7	39.6	40.3
Panel B: Reduced Form						
ΔQuota_m (ρ_1)	1.056**	0.596	0.403	0.463	1.176**	0.948**
	(0.381)	(0.401)	(0.436)	(0.360)	(0.403)	(0.370)
Panel C: Two-Stage Least Squares						
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	3.084†	1.170	0.725	1.192	3.532†	2.364†
	(1.282)	(0.782)	(0.784)	(0.925)	(1.429)	(1.067)
AR 95% CI	[0.742, 6.400]	[-0.460, 2.808]	[-0.896, 2.363]	[-0.749, 3.207]	[0.961, 7.305]	[0.430, 5.155]
Number of Mayors	2,078	2,078	2,078	2,078	2,078	2,078
Observations	4,156	4,156	4,156	4,156	4,156	4,156

Notes: All columns use the fixed 2,078-municipality sample, absorb municipality and year effects, use 2006-population weights, and cluster standard errors by state. The instrument and endogenous beneficiary change are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. All columns start from the complete predetermined control set, interacted with post: incumbent party, 2006 PT presidential vote share, baseline 2008 program intensity and per-family benefit value, log 2006 quota, and municipality and mayor characteristics, including baseline 2008 FPM transfers. The state-by-post column additionally absorbs state-specific changes. GNS (2026) adds the expected quota change under the old allocation rule. Nonlinear adds squares of baseline program intensity, log population, PT vote, and log quota; the last two columns omit the named control. Standard errors are in parentheses. Reduced-form stars use two-sided null-imposed state wild-cluster bootstrap p-values from 9,999 draws: $*\ p < 0.10$, $**\ p < 0.05$, $***\ p < 0.01$. The same star rule applies to first-stage coefficients. † on a 2SLS estimate means its unrounded Anderson–Rubin 95% set excludes zero. Anderson–Rubin 95% confidence sets use quadratic state-clustered inversion with a t_{G-1} critical value and are robust to arbitrary instrument weakness. The CSV reports all three electoral outcomes.

Table 9: Identification Sensitivities 2

	Δ Incumbent's Vote Share						
		Alt. Outcomes		Alt. Shock		Alt. Weights	Alt. Scale
	Main	Winsorized	Registered Votes	Winsorized	Limited to 2009	IPW	Per Families
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: First Stage							
ΔQuota_m (π_1)	0.343***	0.343***	0.343***	0.315***	0.275***	0.340***	0.314***
	(0.053)	(0.053)	(0.053)	(0.044)	(0.039)	(0.050)	(0.047)
Kleibergen–Paap F	41.39	41.39	41.39	51.62	50.31	47.05	44.51
Panel B: Reduced Form							
ΔQuota_m (ρ_1)	1.056**	0.955**	0.950*	1.204**	0.158	0.930**	0.986**
	(0.381)	(0.347)	(0.375)	(0.365)	(0.549)	(0.386)	(0.403)
Panel C: Two-Stage Least Squares							
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	3.084†	2.786†	2.773†	3.826†	0.575	2.738†	3.136†
	(1.282)	(1.157)	(1.242)	(1.254)	(1.973)	(1.207)	(1.487)
AR 95% CI	[0.742, 6.400]	[0.659, 5.760]	[0.487, 5.961]	[1.419, 6.847]	[-4.042, 4.482]	[0.394, 5.635]	[0.454, 7.002]
Bolsa Família Characteristics	✓	✓	✓	✓	✓	✓	✓
Party Characteristics	✓	✓	✓	✓	✓	✓	✓
Municipality Characteristics	✓	✓	✓	✓	✓	✓	✓
Mayor Characteristics	✓	✓	✓	✓	✓	✓	✓
Number of Mayors	2,078	2,078	2,078	2,078	2,078	2,078	2,078
Observations	4,156	4,156	4,156	4,156	4,156	4,156	4,156

Notes: All columns absorb municipality and year effects, use 2006-population weights, and cluster standard errors by state. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD, except the per-family normalization in Column (7). All columns include the complete predetermined control set: 2008 incumbent party, 2006 PT presidential vote share, baseline 2008 program intensity and per-family benefit value, log 2006 quota, and municipality and mayor characteristics, including baseline 2008 FPM transfers. These controls are interacted with post. Standard errors are in parentheses. Column (1) is the preferred valid-vote specification. Columns (2)–(3) winsorize the outcome at the 1st and 99th percentiles or divide incumbent votes by registered voters. Column (4) winsorizes the quota shock. Column (5) replaces the 2012 authorization with the corresponding 2009 authorization and retains the 2008–2012 beneficiary change. Column (6) adds IPW to the population weights. Column (7) expresses the instrument and treatment per 100 Census-2000 families. The winsorized and 2009 quota shocks use their own analysis-weighted SDs. First-stage and reduced-form stars use two-sided null-imposed state wild-cluster bootstrap p-values from 9,999 draws: $*\ p < 0.10$, $**\ p < 0.05$, $***\ p < 0.01$. Anderson–Rubin 95% confidence sets use quadratic state-clustered inversion with a t_{G-1} critical value and are robust to arbitrary instrument weakness. † on a 2SLS estimate means its unrounded AR 95% set excludes zero.

Table 10: Pre-Reform Vote-Share Balance

	Incumbent's Vote Share in 2008	
	(1)	(2)
ΔQuota_m	0.127	-0.070
	(0.254)	(0.228)
Bolsa Família Characteristics	✓	✓
Party Characteristics	✓	✓
Municipality Characteristics		✓
Mayor Characteristics		✓
Number of Mayors	2,078	2,078
Observations	2,078	2,078

Notes: The dependent variable is the vote share of the mayor elected in 2008, measured before the quota reform. Every column is weighted by 2006 population and includes state fixed effects. The quota shock is a rate per 10,000 residents using fixed 2006 population and is divided by its main-sample analysis-weighted SD. Column (1) includes incumbent-party effects and baseline *Bolsa Família* intensity. Column (2) adds municipality and incumbent-mayor characteristics. These controls are baseline and pre-shock: party, program intensity, and mayor characteristics use 2008 values; population uses 2006; and the remaining municipality characteristics use the 2000 Census. Both columns therefore use only pre-shock controls. Standard errors clustered by state are reported in parentheses. Stars use state wild-cluster bootstrap p-values: $*\ p < 0.10$, $**\ p < 0.05$, $***\ p < 0.01$.

Table 11: Selection into Rerunning

	Incumbent Runs Again in 2012	
	(1)	(2)
ΔQuota_m	-2.183	0.045
	(1.156)	(1.017)
Bolsa Família Characteristics	✓	✓
Party Characteristics	✓	✓
Municipality Characteristics		✓
Mayor Characteristics		✓
Number of Mayors	5,492	5,492
Observations	5,492	5,492

Notes: The dependent variable equals 100 if the mayor elected in 2008 appears on the 2012 ballot and zero otherwise. The estimation universe is constructed before conditioning on rerunning. Every column is weighted by 2006 population and includes state fixed effects because municipality and year effects are not identified in this one-observation-per-mayor design. The quota shock is a rate per 10,000 residents using fixed 2006 population and is divided by the main-sample analysis-weighted SD. Column (1) includes incumbent-party effects and baseline *Bolsa Família* intensity. Column (2) adds municipality and incumbent-mayor characteristics. These controls are baseline and pre-shock: party, program intensity, and mayor characteristics use 2008 values; population uses 2006; and the remaining municipality characteristics use the 2000 Census. Both columns therefore use only pre-shock controls. Standard errors clustered by state are reported in parentheses. Stars use state wild-cluster bootstrap p-values: $*\ p < 0.10$, $**\ p < 0.05$, $***\ p < 0.01$.

Table 12: Heterogeneity by Municipality Size

	Δ Incumbent's Vote Share		Δ Reelection Probability		Δ Turnout	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Below Median						
ΔQuota_m (ρ_1)	-0.077	-0.013	-1.090	-0.823	-0.253**	-0.246*
	(0.321)	(0.304)	(0.867)	(0.809)	(0.131)	(0.122)
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	-0.174	-0.027	-2.465	-1.765	-0.572	-0.528
	(0.741)	(0.653)	(1.933)	(1.684)	(0.332)	(0.284)
AR 95% CI	[-2.171, 1.167]	[-1.597, 1.258]	[-6.811, 1.727]	[-5.296, 2.031]	[-1.459, 0.034]	[-1.231, 0.013]
Panel B: Above Median						
ΔQuota_m (ρ_1)	2.386**	1.531**	3.693	2.879	-0.474***	-0.378
	(0.706)	(0.483)	(2.864)	(1.858)	(0.224)	(0.253)
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	11.642†	5.077†	18.018	9.547	-2.314†	-1.254
	(4.282)	(1.827)	(13.068)	(5.999)	(1.261)	(0.879)
AR 95% CI	[4.247, 24.455]	[1.690, 9.650]	[-13.265, 46.232]	[-3.406, 22.443]	[-5.954, -0.055]	[-3.340, 0.466]
Panel C: Below the 75th Percentile						
ΔQuota_m (ρ_1)	0.031	0.014	-0.628	-0.587	-0.374**	-0.306*
	(0.278)	(0.298)	(0.797)	(0.928)	(0.211)	(0.168)
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	0.087	0.034	-1.778	-1.460	-1.060	-0.760
	(0.782)	(0.740)	(2.269)	(2.301)	(0.680)	(0.458)
AR 95% CI	[-1.791, 1.648]	[-1.620, 1.574]	[-6.951, 2.991]	[-6.463, 3.470]	[-2.893, 0.150]	[-1.902, 0.095]
Panel D: Above the 75th Percentile						
ΔQuota_m (ρ_1)	3.832***	2.769***	5.774	4.484	-0.448**	-0.399
	(0.940)	(0.739)	(4.237)	(2.338)	(0.217)	(0.290)
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	18.762†	9.670†	28.274	15.658	-2.195†	-1.393
	(6.232)	(3.037)	(19.273)	(8.027)	(1.264)	(1.105)
AR 95% CI	[8.612, 40.197]	[4.219, 18.043]	[-18.672, 72.604]	[-1.271, 34.493]	[-6.282, -0.007]	[-4.360, 0.644]
Panel E: Above the 90th Percentile						
ΔQuota_m (ρ_1)	6.982***	5.353***	9.194	6.459	-0.421	-0.419
	(1.132)	(1.276)	(5.972)	(4.107)	(0.244)	(0.346)
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	28.742†	17.229†	37.847	20.788	-1.733	-1.347
	(12.631)	(5.530)	(16.649)	(10.112)	(1.554)	(1.340)
AR 95% CI	[14.376, 503.851]	[8.405, 39.391]	[-88.942, 131.380]	[-11.681, 40.926]	[-53.730, 0.253]	[-6.758, 0.774]
Bolsa Família Characteristics	✓	✓	✓	✓	✓	✓
Party Characteristics	✓	✓	✓	✓	✓	✓
Municipality Characteristics		✓		✓		✓
Mayor Characteristics		✓		✓		✓

Notes: All columns absorb municipality and year effects. All columns use 2006-population weights and state-clustered standard errors. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. Column (1) includes 2008 incumbent party, 2006 PT presidential vote share, baseline 2008 program intensity and per-family benefit value, and log 2006 quota. Column (2) adds municipality and mayor characteristics, including baseline 2008 FPM transfers in the municipality group. Predetermined controls are interacted with post. Standard errors are in parentheses. The Column (1)–(2) control sequence repeats in Columns (3)–(4) and (5)–(6). Reelection probability is conditional on the incumbent rerunning in 2012. Anderson–Rubin 95% confidence sets use quadratic state-clustered inversion with a t_{G-1} critical value and are robust to arbitrary instrument weakness. Reduced-form stars use two-sided null-imposed state wild-cluster bootstrap p-values from 9,999 draws: $*\ p < 0.10$, $**\ p < 0.05$, $***\ p < 0.01$. † on a 2SLS estimate means its unrounded AR 95% set excludes zero. Panels use the 2006-population median, 75th percentile, and 90th percentile calculated in the headline sample. Below-median is nested within below-75th; above-90th is nested within above-75th, which is nested within above-median. Models are estimated separately within each displayed subsample; the panels are not mutually exclusive.

Table 13: Heterogeneity by Subgroup: Frey Sample and Baseline *Bolsa Família* Intensity

	Frey 2019			Baseline *Bolsa Família* Intensity		
	Outside Sample	Inside Sample	Δ	Below Median	Above Median	Δ
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: First Stage						
ΔQuota_m (π_1)	0.235***	0.383***	-0.148**	0.246***	0.390***	-0.144*
	(0.056)	(0.060)	(0.057)	(0.046)	(0.062)	(0.060)
Conditional First-Stage F	17.83	40.67		28.86	39.11	
Panel B: Reduced Form						
ΔQuota_m (ρ_1)	3.021***	0.022	2.999**	2.855**	0.016	2.839*
	(0.823)	(0.286)	(0.917)	(1.057)	(0.272)	(1.196)
Panel C: Two-Stage Least Squares						
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	12.836†	0.057	12.778	11.597†	0.042	11.555
	(4.168)	(0.746)	(4.401)	(4.461)	(0.694)	(4.869)
Conditional AR 95% CI	[5.639, 26.269]	[-1.597, 1.650]		[2.877, 22.948]	[-1.676, 1.377]	
Bolsa Família Characteristics	✓	✓		✓	✓	
Party Characteristics	✓	✓		✓	✓	
Municipality Characteristics	✓	✓		✓	✓	
Mayor Characteristics	✓	✓		✓	✓	
Number of Mayors	834	1,244	2,078	1,039	1,039	2,078
Observations	4,156	4,156	4,156	4,156	4,156	4,156

Notes: All estimated subgroup coefficients absorb municipality and year effects. All use 2006-population weights and state-clustered standard errors. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one full-sample analysis-weighted SD. Each subgroup includes 2008 incumbent party, 2006 PT presidential vote share, baseline 2008 program intensity and per-family benefit value, log 2006 quota, and municipality and mayor characteristics, including baseline 2008 FPM transfers in the municipality group. Predetermined controls are interacted with post. Standard errors are in parentheses. Each three-column block uses the full headline sample. The Frey sample comprises municipalities with 2000 population below 30,000 and 2010 HDI below 0.7. Each block is one fully interacted dual-IV regression. The instrument, endogenous variable, and complete control set are allowed to vary across the complementary subgroup indicators; Δ is the first coefficient minus the second. The saturated system reproduces separate subgroup coefficients while retaining one joint model for the displayed comparison. Number of Mayors counts each subgroup (and the full block for the contrast); Observations reports the full sample used by the saturated regression in every column. Contrast columns are calculated from the joint model and therefore do not receive control check marks. First-stage and reduced-form stars use two-sided null-imposed state wild-cluster bootstrap p-values from 9,999 draws, including direct tests of the displayed contrasts: $*\ p < 0.10$, $**\ p < 0.05$, $***\ p < 0.01$. Conditional first-stage F statistics follow Sanderson and Windmeijer (2016). Group AR 95% sets use quadratic state-clustered inversion with a t_{G-1} critical value, conditional on the other group coefficient at its 2SLS estimate. They are not exact joint weak-IV confidence regions. † attached to a subgroup 2SLS estimate denotes that its conditional AR 95% set excludes zero; contrasts have no AR marker.

Table 14: Heterogeneity by Subgroup: Northeast States and PT Incumbent

	Northeast States			PT Incumbent		
	Other Regions	Northeast	Δ	Other Parties	PT	Δ
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: First Stage						
ΔQuota_m (π_1)	0.416***	0.504***	-0.088	0.325***	0.390***	-0.065
	(0.078)	(0.029)	(0.083)	(0.040)	(0.125)	(0.106)
Conditional First-Stage F	28.67	302.73		54.61	6.86	
Panel B: Reduced Form						
ΔQuota_m (ρ_1)	1.059	0.618	0.440	1.431**	-0.742	2.173
	(0.801)	(0.532)	(0.961)	(0.528)	(1.021)	(1.428)
Panel C: Two-Stage Least Squares						
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	2.545	1.227	1.318	4.404†	-1.903	6.307
	(2.118)	(1.029)	(2.355)	(1.657)	(2.755)	(4.157)
Conditional AR 95% CI	[-1.296, 8.357]	[-0.998, 3.275]		[1.054, 8.303]	[-84.885, 0.750]	
Bolsa Família Characteristics	✓	✓		✓	✓	
Party Characteristics	✓	✓		✓	✓	
Municipality Characteristics	✓	✓		✓	✓	
Mayor Characteristics	✓	✓		✓	✓	
Number of Mayors	1,373	705	2,078	1,821	257	2,078
Observations	4,156	4,156	4,156	4,156	4,156	4,156

Notes: All estimated subgroup coefficients absorb municipality and year effects. All use 2006-population weights and state-clustered standard errors. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one full-sample analysis-weighted SD. Each subgroup includes 2008 incumbent party, 2006 PT presidential vote share, baseline 2008 program intensity and per-family benefit value, log 2006 quota, and municipality and mayor characteristics, including baseline 2008 FPM transfers in the municipality group. Predetermined controls are interacted with post. Standard errors are in parentheses. Each three-column block uses the full headline sample. Each block is one fully interacted dual-IV regression. The instrument, endogenous variable, and complete control set are allowed to vary across the complementary subgroup indicators; Δ is the first coefficient minus the second. The saturated system reproduces separate subgroup coefficients while retaining one joint model for the displayed comparison. Number of Mayors counts each subgroup (and the full block for the contrast); Observations reports the full sample used by the saturated regression in every column. Contrast columns are calculated from the joint model and therefore do not receive control check marks. First-stage and reduced-form stars use two-sided null-imposed state wild-cluster bootstrap p-values from 9,999 draws, including direct tests of the displayed contrasts: $*$ $p < 0.10$, $**$ $p < 0.05$, $***$ $p < 0.01$. Conditional first-stage F statistics follow Sanderson and Windmeijer (2016). Group AR 95% sets use quadratic state-clustered inversion with a t_{G-1} critical value, conditional on the other group coefficient at its 2SLS estimate. They are not exact joint weak-IV confidence regions. † attached to a subgroup 2SLS estimate denotes that its conditional AR 95% set excludes zero; contrasts have no AR marker.

Table 15: Continuous-Quota and Discrete-Beneficiary Specifications

	Δ Incumbent's Vote Share		Δ Reelection Probability		Δ Turnout	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Reduced Form						
ΔQuota_m (ρ_1)	1.863**	1.056**	3.103	1.409	-0.398***	-0.328*
	(0.571)	(0.381)	(2.056)	(1.443)	(0.195)	(0.196)
Panel B: Continuous						
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	7.395†	3.084†	12.315	4.114	-1.578	-0.956
	(2.932)	(1.282)	(8.159)	(4.233)	(0.896)	(0.610)
AR 95% CI	[2.442, 16.315]	[0.742, 6.400]	[-4.933, 32.154]	[-4.751, 13.665]	[-4.167, 0.015]	[-2.458, 0.214]
Panel C: Above Median						
$\mathbf{1}\{\Delta\widehat{\text{Beneficiaries}}_m \geq c_D\}$ (β_1)	45.830†	12.812†	76.324	17.094	-9.783	-3.972
	(25.043)	(5.048)	(49.149)	(17.338)	(6.376)	(2.255)
AR 95% CI	[13.989, 615.434]	[3.215, 25.235]	[-57.161, 599.651]	[-20.272, 54.724]	[-128.712, 0.102]	[-8.725, 1.032]
Panel D: Above the 75th Percentile						
$\mathbf{1}\{\Delta\widehat{\text{Beneficiaries}}_m \geq c_D\}$ (β_1)	29.396†	11.484†	48.955	15.321	-6.275	-3.560
	(10.387)	(4.207)	(29.635)	(15.231)	(3.489)	(2.268)
AR 95% CI	[10.721, 59.469]	[3.110, 21.736]	[-22.472, 112.960]	[-19.342, 47.779]	[-16.280, 0.060]	[-9.268, 0.809]
Panel E: Above the 90th Percentile						
$\mathbf{1}\{\Delta\widehat{\text{Beneficiaries}}_m \geq c_D\}$ (β_1)	32.083†	15.472†	53.430	20.642	-6.848	-4.797
	(12.575)	(6.524)	(36.066)	(21.356)	(4.079)	(3.361)
AR 95% CI	[10.862, 72.058]	[3.852, 34.987]	[-21.210, 146.921]	[-24.903, 73.986]	[-19.766, 0.060]	[-15.209, 0.983]
Bolsa Família Characteristics	✓	✓	✓	✓	✓	✓
Party Characteristics	✓	✓	✓	✓	✓	✓
Municipality Characteristics		✓		✓		✓
Mayor Characteristics		✓		✓		✓
Number of Mayors	2,078	2,078	2,078	2,078	2,078	2,078
Observations	4,156	4,156	4,156	4,156	4,156	4,156

Notes: All columns absorb municipality and year effects, use 2006-population weights, and cluster standard errors by state. The quota instrument and continuous beneficiary change are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. Columns (1), (3), and (5) include 2008 incumbent party, 2006 PT presidential vote share, baseline 2008 program intensity and per-family benefit value, and log 2006 quota. Columns (2), (4), and (6) add municipality and mayor characteristics, including baseline 2008 FPM transfers in the municipality group. Predetermined controls are interacted with post. Standard errors are in parentheses. The quota shock remains continuous in every panel. Panel B retains continuous beneficiary change; Panels C–E use an indicator that the 2008–2012 change meets or exceeds the headline-sample median, 75th percentile, or 90th percentile, respectively. Their 2SLS coefficients describe crossing the displayed threshold, not a continuous dose response. The reduced form is common across panels and shown once. Reelection probability is conditional on the incumbent rerunning in 2012. Anderson–Rubin 95% confidence sets use quadratic state-clustered inversion with a t_{G-1} critical value and are robust to arbitrary instrument weakness. Reduced-form stars use two-sided null-imposed state wild-cluster bootstrap p-values from 9,999 draws: $*\ p < 0.10$, $**\ p < 0.05$, $***\ p < 0.01$. † on a 2SLS estimate means its unrounded AR 95% set excludes zero.

Table 16: Heterogeneity by CRAS Presence

	Δ Incumbent's Vote Share			Δ Reelection Probability			Δ Turnout		
	Without CRAS	With CRAS	Δ	Without CRAS	With CRAS	Δ	Without CRAS	With CRAS	Δ
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: First Stage									
ΔQuota_m (π_1)	0.352	0.344	0.007	0.352	0.344	0.007	0.352	0.344	0.007
	(0.047)	(0.061)	(0.059)	(0.047)	(0.061)	(0.059)	(0.047)	(0.061)	(0.059)
Conditional First-Stage F	56.78	31.60		56.78	31.60		56.78	31.60	
Panel B: Reduced Form									
ΔQuota_m (ρ_1)	-0.345	1.411	-1.756	1.797	1.819	-0.022	-0.118	-0.353	0.235
	(0.583)	(0.437)	(0.703)	(0.984)	(1.682)	(1.999)	(0.081)	(0.229)	(0.212)
Panel C: Two-Stage Least Squares									
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	-0.980	4.098†	-5.078	5.109	5.281	-0.172	-0.335	-1.025	0.689
	(1.667)	(1.560)	(2.110)	(2.799)	(4.952)	(5.680)	(0.232)	(0.711)	(0.666)
Conditional AR 95% CI	[-4.653, 2.488]	[1.361, 8.454]		[-0.679, 11.315]	[-4.989, 16.980]		[-0.850, 0.144]	[-2.849, 0.333]	
Bolsa Família Characteristics	✓	✓		✓	✓		✓	✓	
Party Characteristics	✓	✓		✓	✓		✓	✓	
Municipality Characteristics	✓	✓		✓	✓		✓	✓	
Mayor Characteristics	✓	✓		✓	✓		✓	✓	
Number of Mayors	2,078	2,078	2,078	2,078	2,078	2,078	2,078	2,078	2,078
Observations	4,156	4,156	4,156	4,156	4,156	4,156	4,156	4,156	4,156

Notes: All columns absorb municipality and year fixed effects. All columns use 2006-population weights and state-clustered standard errors. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. Party characteristics include 2008 incumbent party and 2006 PT presidential vote share. *Bolsa Família* characteristics include baseline 2008 program intensity, per-family benefit value, and log 2006 quota. Municipality characteristics include baseline 2008 *FPM* transfers. Predetermined controls are interacted with post. Standard errors are in parentheses. Reelection probability is conditional on the incumbent rerunning in 2012. † on a subgroup 2SLS estimate denotes exclusion of zero by its unrounded conditional AR 95% set; contrasts have no marker. Each outcome is estimated in one fully interacted dual-IV system allowing the standardized instrument, treatment, and complete control set to vary with 2008 *CRAS* presence and its complement; Δ is the first coefficient minus the second. The saturated system reproduces separate subgroup coefficients while retaining one joint model for the displayed comparison. Conditional first-stage F statistics follow Sanderson and Windmeijer (2016). AR sets are plug-in conditional diagnostics, not exact joint weak-IV regions.

Table 17: Descriptive Statistics by CRAS Presence

Municipalities With CRAS	N	Min	Mean	Median	Max	SD	**Municipalities Without CRAS**	N	Min	Mean	Median	Max	SD
Panel A: Census Information							**Panel A:** Census Information						
Population (2000)	1,394	1,262.00	36,787.28	13,389.00	5,773,130.00	177,650.70	Population (2000)	684	795.00	10,064.50	6,421.50	122,377.00	11,514.12
Urban Population (2000) (%)	1,394	4.67	59.52	59.54	100.00	23.56	Urban Population (2000) (%)	684	0.00	57.33	59.09	100.00	22.77
Poor Population (2000) (%)	1,394	7.66	67.53	74.39	99.00	20.71	Poor Population (2000) (%)	684	10.91	57.07	56.78	96.97	19.79
Income per capita (2000) (R$)	1,394	69.20	315.16	243.12	1,235.29	197.96	Income per capita (2000) (R$)	684	76.77	382.74	368.90	1,759.76	182.02
Gini (2000) (%)	1,394	35.00	55.79	56.00	79.00	6.26	Gini (2000) (%)	684	30.00	52.92	53.00	80.00	7.22
Formal Workers (2000) (%)	1,394	2.89	33.69	30.49	80.88	18.25	Formal Workers (2000) (%)	684	4.74	40.09	39.79	86.38	17.69
Panel B: *Bolsa Família* Program Information							**Panel B:** *Bolsa Família* Program Information						
Permitted *Bolsa Família* Families (2008)	1,394	58.00	2,485.34	1,351.00	164,913.00	6,090.08	Permitted *Bolsa Família* Families (2008)	684	24.00	702.24	457.00	6,344.00	782.52
Permitted *Bolsa Família* Families (2008) (%)	1,394	0.82	9.56	9.48	30.25	4.70	Permitted *Bolsa Família* Families (2008) (%)	684	0.77	7.51	6.97	33.00	4.41
Permitted *Bolsa Família* Families (2008) per 10,000 Residents	1,394	81.88	956.42	948.41	3,025.26	469.94	Permitted *Bolsa Família* Families (2008) per 10,000 Residents	684	77.08	751.21	696.50	3,299.89	441.17
Δ Permitted *Bolsa Família* Families per 10,000 Residents	1,394	-1,000.62	149.99	126.26	2,686.38	267.25	Δ Permitted *Bolsa Família* Families per 10,000 Residents	684	-889.25	60.15	26.58	2,865.11	287.54
Realized *Bolsa Família* Families (2008)	1,394	23.67	2,455.52	1,339.92	133,322.75	5,424.96	Realized *Bolsa Família* Families (2008)	684	5.50	680.56	425.79	6,362.42	792.05
Realized *Bolsa Família* Families (2008) (%)	1,394	0.82	9.65	9.80	24.91	4.88	Realized *Bolsa Família* Families (2008) (%)	684	0.26	7.07	6.32	29.64	4.48
Realized *Bolsa Família* Families (2008) per 10,000 Residents	1,394	82.11	964.74	979.67	2,491.38	488.26	Realized *Bolsa Família* Families (2008) per 10,000 Residents	684	25.81	706.89	631.86	2,963.95	447.92
Δ Realized *Bolsa Família* Families per 10,000 Residents	1,394	-398.59	230.89	203.79	2,979.12	235.14	Δ Realized *Bolsa Família* Families per 10,000 Residents	684	-301.46	147.68	95.65	2,440.54	235.35
Panel C: Election Information							**Panel C:** Election Information						
Candidates (2008)	1,394	2.00	2.90	3.00	12.00	1.16	Candidates (2008)	684	2.00	2.59	2.00	7.00	0.81
Candidates (2012)	1,394	2.00	2.82	3.00	9.00	1.07	Candidates (2012)	684	2.00	2.44	2.00	8.00	0.78
Seeking Reelection (2012) (%)	1,394	100.00	100.00	100.00	100.00	0.00	Seeking Reelection (2012) (%)	684	100.00	100.00	100.00	100.00	0.00
Winner's Vote Share (2008) (%)	1,394	24.06	52.90	52.60	99.16	9.91	Winner's Vote Share (2008) (%)	684	28.05	52.97	52.98	98.18	8.37
Incumbent Vote Share (2012) (%)	1,394	0.18	49.82	49.22	100.00	14.64	Incumbent Vote Share (2012) (%)	684	5.05	50.33	50.61	100.00	12.70
Δ Incumbent Vote Share (pp)	1,394	-71.82	-3.08	-3.25	54.95	15.83	Δ Incumbent Vote Share (pp)	684	-43.28	-2.64	-2.40	55.47	13.62
Panel D: Candidate Information							**Panel D:** Candidate Information						
Female Mayor (2008) (%)	1,394	0.00	11.33	0.00	100.00	31.71	Female Mayor (2008) (%)	684	0.00	7.75	0.00	100.00	26.76
High School or More (2008) (%)	1,394	0.00	83.07	100.00	100.00	37.51	High School or More (2008) (%)	684	0.00	77.19	100.00	100.00	41.99
College Degree (2008) (%)	1,394	0.00	48.78	0.00	100.00	50.00	College Degree (2008) (%)	684	0.00	41.52	0.00	100.00	49.31
Mayor Age (2008)	1,394	21.00	46.57	46.00	78.00	9.27	Mayor Age (2008)	684	23.00	46.38	46.00	81.00	8.68
Married (2008) (%)	1,394	0.00	75.75	100.00	100.00	42.87	Married (2008) (%)	684	0.00	79.68	100.00	100.00	40.27
Public-Sector Occupation (2008) (%)	1,394	0.00	15.93	0.00	100.00	36.60	Public-Sector Occupation (2008) (%)	684	0.00	15.35	0.00	100.00	36.07
Agricultural Occupation (2008) (%)	1,394	0.00	13.70	0.00	100.00	34.40	Agricultural Occupation (2008) (%)	684	0.00	16.08	0.00	100.00	36.76
Business Occupation (2008) (%)	1,394	0.00	25.11	0.00	100.00	43.38	Business Occupation (2008) (%)	684	0.00	28.80	0.00	100.00	45.32

Notes: The two municipality groups partition the locked headline sample by CRAS presence before the reform. The four information panels repeat for both groups. Variables, scaling, and unweighted summary statistics match the main descriptive table; N counts nonmissing municipalities. The table includes baseline characteristics and the explicitly dated program and electoral outcomes shown in that table. Differences are descriptive, not estimates of the effect of CRAS.

Table 18: Enrollment and Electoral Responses by Baseline CRAS

	No CRAS (1)	CRAS (2)	Difference (3)
Panel A: Enrollment per 10,000 residents			
Population weights			
Quota response	63.772***	62.453***	-1.320
State-clustered s.e.	(8.463)	(11.110)	(10.784)
Wild-cluster p	< 0.001	< 0.001	0.907
Holm-adjusted p	< 0.001	< 0.001	0.907
Overlap weights			
Quota response	64.107***	73.315***	9.207
State-clustered s.e.	(8.487)	(11.724)	(9.405)
Wild-cluster p	< 0.001	< 0.001	0.391
Holm-adjusted p	< 0.001	< 0.001	1.000
Panel B: Incumbent vote share (percentage points)			
Population weights			
Quota response	-0.345	1.411***	1.756*
State-clustered s.e.	(0.583)	(0.437)	(0.703)
Wild-cluster p	0.610	0.004	0.054
Holm-adjusted p	0.610	0.008	0.162
Overlap weights			
Quota response	-0.250	0.016	0.266
State-clustered s.e.	(0.655)	(0.378)	(0.750)
Wild-cluster p	0.720	0.968	0.740
Holm-adjusted p	0.720	0.968	1.000
Number of Mayors	684	1,394	2,078
Observations	4,156	4,156	4,156

Notes: Each response is per one full-sample 2006-population-weighted SD of the quota shock. Enrollment is beneficiary families per 10,000 residents; the electoral outcome is the change in incumbent vote share in percentage points. The fully interacted specification retains the same 2,078 municipalities and 26 state clusters in both weighting schemes. Overlap weights balance seven observed baseline characteristics and state shares. Parentheses give state-clustered standard errors. Stars use two-sided null-imposed state wild-cluster tests with 9,999 draws: $*p < 0.10$, $**\ p < 0.05$, $***p < 0.01$. Holm-adjusted p-values account for three outcomes within each weighting scheme and coefficient column. Inference conditions on fitted weights. Balancing observed covariates does not identify the causal effect of an office or eliminate unobserved differences.

Table 19: Heterogeneity by CRAS Intensity: Active Outreach

	Δ Incumbent's Vote Share			Δ Reelection Probability			Δ Turnout		
	Without Active Outreach	With Active Outreach	Δ	Without Active Outreach	With Active Outreach	Δ	Without Active Outreach	With Active Outreach	Δ
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: First Stage									
ΔQuota_m (π_1)	0.479	0.315	0.164	0.479	0.315	0.164	0.479	0.315	0.164
	(0.109)	(0.042)	(0.085)	(0.109)	(0.042)	(0.085)	(0.109)	(0.042)	(0.085)
Conditional First-Stage F	19.32	56.40		19.32	56.40		19.32	56.40	
Panel B: Reduced Form									
ΔQuota_m (ρ_1)	0.250	1.617	-1.366	-2.342	3.083	-5.425	-0.208	-0.357	0.148
	(0.688)	(0.613)	(0.981)	(1.375)	(2.207)	(2.333)	(0.179)	(0.254)	(0.151)
Panel C: Two-Stage Least Squares									
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	0.523	5.135†	-4.612	-4.892	9.791	-14.683	-0.435	-1.133	0.698
	(1.437)	(1.993)	(2.695)	(2.752)	(6.768)	(6.760)	(0.406)	(0.831)	(0.561)
Conditional AR 95% CI	[-2.739, 3.965]	[1.147, 9.704]		[-11.629, 1.218]	[-5.094, 23.907]		[-1.616, 0.323]	[-3.046, 0.522]	
Bolsa Família Characteristics	✓	✓		✓	✓		✓	✓	
Party Characteristics	✓	✓		✓	✓		✓	✓	
Municipality Characteristics	✓	✓		✓	✓		✓	✓	
Mayor Characteristics	✓	✓		✓	✓		✓	✓	
Number of Mayors	1,394	1,394	1,394	1,394	1,394	1,394	1,394	1,394	1,394
Observations	2,788	2,788	2,788	2,788	2,788	2,788	2,788	2,788	2,788

Notes: All columns absorb municipality and year fixed effects. All columns use 2006-population weights and state-clustered standard errors. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. Party characteristics include 2008 incumbent party and 2006 PT presidential vote share. *Bolsa Família* characteristics include baseline 2008 program intensity, per-family benefit value, and log 2006 quota. Municipality characteristics include baseline 2008 *FPM* transfers. Predetermined controls are interacted with post. Standard errors are in parentheses. Reelection probability is conditional on the incumbent rerunning in 2012. † on a subgroup 2SLS estimate denotes exclusion of zero by its unrounded conditional AR 95% set; contrasts have no marker. Each outcome is estimated within 2008 *CRAS* municipalities in one fully interacted dual-IV system comparing units with and without active outreach; Δ is Without Active Outreach minus With Active Outreach. The instrument, treatment, and complete control set can vary across the two groups. Conditional first-stage F statistics follow Sanderson and Windmeijer (2016). AR sets are plug-in conditional diagnostics, not exact joint weak-IV regions. Groups are predetermined scope conditions, not causal effects of adopting outreach.

Table 20: Heterogeneity by CRAS Intensity: *Proteção e Atendimento Integral à Família* (PAIF)

	Δ Incumbent's Vote Share			Δ Reelection Probability			Δ Turnout		
	Without PAIF	With PAIF	Δ	Without PAIF	With PAIF	Δ	Without PAIF	With PAIF	Δ
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: First Stage									
ΔQuota_m (π_1)	0.409	0.335	0.074	0.409	0.335	0.074	0.409	0.335	0.074
	(0.133)	(0.052)	(0.120)	(0.133)	(0.052)	(0.120)	(0.133)	(0.052)	(0.120)
Conditional First-Stage F	9.53	41.70		9.53	41.70		9.53	41.70	
Panel B: Reduced Form									
ΔQuota_m (ρ_1)	0.687	1.625	-0.938	-1.050	2.781	-3.831	-0.471	-0.359	-0.112
	(0.697)	(0.709)	(1.178)	(1.770)	(2.580)	(3.755)	(0.286)	(0.219)	(0.166)
Panel C: Two-Stage Least Squares									
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	1.677	4.846†	-3.169	-2.564	8.296	-10.860	-1.150	-1.071	-0.079
	(1.779)	(2.275)	(3.361)	(4.513)	(7.595)	(10.697)	(0.871)	(0.698)	(0.598)
Conditional AR 95% CI	[-2.132, 8.026]	[0.484, 10.448]		[-19.065, 6.870]	[-8.172, 24.836]		[-5.281, 0.253]	[-2.793, 0.266]	
Bolsa Família Characteristics	✓	✓		✓	✓		✓	✓	
Party Characteristics	✓	✓		✓	✓		✓	✓	
Municipality Characteristics	✓	✓		✓	✓		✓	✓	
Mayor Characteristics	✓	✓		✓	✓		✓	✓	
Number of Mayors	1,394	1,394	1,394	1,394	1,394	1,394	1,394	1,394	1,394
Observations	2,788	2,788	2,788	2,788	2,788	2,788	2,788	2,788	2,788

Notes: All columns absorb municipality and year fixed effects. All columns use 2006-population weights and state-clustered standard errors. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. Party characteristics include 2008 incumbent party and 2006 PT presidential vote share. *Bolsa Família* characteristics include baseline 2008 program intensity, per-family benefit value, and log 2006 quota. Municipality characteristics include baseline 2008 *FPM* transfers. Predetermined controls are interacted with post. Standard errors are in parentheses. Reelection probability is conditional on the incumbent rerunning in 2012. † on a subgroup 2SLS estimate denotes exclusion of zero by its unrounded conditional AR 95% set; contrasts have no marker. Each outcome is estimated within 2008 *CRAS* municipalities in one fully interacted dual-IV system comparing units with and without PAIF-tracked *Bolsa Família* families; Δ is Without PAIF minus With PAIF. The instrument, treatment, and complete control set can vary across the two groups. Conditional first-stage F statistics follow Sanderson and Windmeijer (2016). AR sets are plug-in conditional diagnostics, not exact joint weak-IV regions. Groups are predetermined scope conditions, not causal effects of adopting PAIF services.

Table 21: Heterogeneity by CRAS Intensity: *Proteção e Atendimento Integral à Família* (PAIF) Volume

	Δ Incumbent's Vote Share			Δ Reelection Probability			Δ Turnout		
	Below Median	Above Median	Δ	Below Median	Above Median	Δ	Below Median	Above Median	Δ
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: First Stage									
ΔQuota_m (π_1)	0.355	0.330	0.026	0.355	0.330	0.026	0.355	0.330	0.026
	(0.065)	(0.048)	(0.071)	(0.065)	(0.048)	(0.071)	(0.065)	(0.048)	(0.071)
Conditional First-Stage F	30.32	46.17		30.32	46.17		30.32	46.17	
Panel B: Reduced Form									
ΔQuota_m (ρ_1)	-0.702	4.252	-4.954	-4.502	7.990	-12.492	-0.342	-0.318	-0.025
	(0.913)	(1.517)	(2.118)	(2.430)	(3.703)	(4.183)	(0.334)	(0.150)	(0.318)
Panel C: Two-Stage Least Squares									
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	-1.976	12.905†	-14.881	-12.667	24.248†	-36.915	-0.963	-0.964†	0.001
	(2.560)	(5.333)	(7.016)	(6.544)	(11.008)	(12.561)	(0.965)	(0.553)	(0.991)
Conditional AR 95% CI	[-7.772, 3.601]	[3.177, 26.560]		[-27.467, 1.602]	[1.177, 48.781]		[-3.313, 0.990]	[-2.479, -0.022]	
Bolsa Família Characteristics	✓	✓		✓	✓		✓	✓	
Party Characteristics	✓	✓		✓	✓		✓	✓	
Municipality Characteristics	✓	✓		✓	✓		✓	✓	
Mayor Characteristics	✓	✓		✓	✓		✓	✓	
Number of Mayors	962	962	962	962	962	962	962	962	962
Observations	1,924	1,924	1,924	1,924	1,924	1,924	1,924	1,924	1,924

Notes: All columns absorb municipality and year fixed effects. All columns use 2006-population weights and state-clustered standard errors. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. Party characteristics include 2008 incumbent party and 2006 PT presidential vote share. *Bolsa Família* characteristics include baseline 2008 program intensity, per-family benefit value, and log 2006 quota. Municipality characteristics include baseline 2008 *FPM* transfers. Predetermined controls are interacted with post. Standard errors are in parentheses. Reelection probability is conditional on the incumbent rerunning in 2012. † on a subgroup 2SLS estimate denotes exclusion of zero by its unrounded conditional AR 95% set; contrasts have no marker. The sample contains 2008 *CRAS* municipalities reporting a positive number of *Bolsa Família* families followed through PAIF. Low and high volume are defined relative to the median positive family count in the headline sample; Δ is Below Median minus Above Median. The fully interacted dual-IV system allows the instrument, treatment, and complete control set to vary across the two groups. Conditional first-stage F statistics follow Sanderson and Windmeijer (2016).

Table 22: Electoral Responses by CRAS Age Before the Reform

	Δ Incumbent's Vote Share			Δ Reelection Probability			Δ Turnout		
	w/o CRAS	Opened by 2006	Opened 2007–2008	w/o CRAS	Opened by 2006	Opened 2007–2008	w/o CRAS	Opened by 2006	Opened 2007–2008
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: First Stage									
ΔQuota_m (π_1)	0.352***	0.312***	0.453***	0.352***	0.312***	0.453***	0.352***	0.312***	0.453***
	(0.047)	(0.043)	(0.095)	(0.047)	(0.043)	(0.095)	(0.047)	(0.043)	(0.095)
Kleibergen–Paap F	55.79	53.97	22.69	55.79	53.97	22.69	55.79	53.97	22.69
Panel B: Reduced Form									
ΔQuota_m (ρ_1)	-0.345	1.907***	1.141**	1.797	2.847	0.264	-0.118	-0.299	-0.467***
	(0.588)	(0.626)	(0.494)	(0.993)	(2.421)	(1.931)	(0.082)	(0.260)	(0.180)
Panel C: Two-Stage Least Squares									
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	-0.980	6.106†	2.520†	5.109	9.115	0.582	-0.335	-0.959	-1.032†
	(1.682)	(2.095)	(1.361)	(2.824)	(7.517)	(4.312)	(0.234)	(0.858)	(0.519)
AR 95% CI	[-4.73, 2.56]	[1.99, 11.02]	[0.23, 6.77]	[-0.79, 11.45]	[-7.54, 24.74]	[-7.58, 12.52]	[-0.86, 0.15]	[-2.94, 0.75]	[-2.69, -0.18]
Bolsa Família Characteristics	✓	✓	✓	✓	✓	✓	✓	✓	✓
Party Characteristics	✓	✓	✓	✓	✓	✓	✓	✓	✓
Municipality Characteristics	✓	✓	✓	✓	✓	✓	✓	✓	✓
Mayor Characteristics	✓	✓	✓	✓	✓	✓	✓	✓	✓
Number of Mayors	684	831	563	684	831	563	684	831	563
Observations	1,368	1,662	1,126	1,368	1,662	1,126	1,368	1,662	1,126

Notes: All columns absorb municipality and year fixed effects. All columns use 2006-population weights and state-clustered standard errors. The quota instrument and beneficiary-family expansion are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. *Bolsa Família* characteristics include baseline 2008 program intensity, per-family benefit value, and log 2006 quota. Party characteristics include 2008 incumbent party and 2006 PT presidential vote share. Municipality characteristics include baseline 2008 FPM transfers. Predetermined controls are interacted with post. Standard errors are in parentheses. The three CRAS-age groups are estimated separately. Opening years come from the baseline census; missing dates are excluded. w/o CRAS denotes no baseline census record, not verified absence of an office. The reported Kleibergen–Paap F is the just-identified excluded-instrument state-clustered Wald statistic. First-stage and reduced-form stars use null-imposed state wild-cluster bootstrap p-values from 9,999 draws: $*\ p < 0.10$, $**\ p < 0.05$, $***\ p < 0.01$. Anderson–Rubin 95% confidence sets use quadratic state-clustered inversion with a t_{G-1} critical value and are robust to arbitrary instrument weakness. † on a 2SLS estimate denotes exclusion of zero by its unrounded AR set. Disconnected two-tail sets use their conservative full-real-line envelope; exact components remain in the result files. Reelection probability is conditional on the incumbent rerunning in 2012. Subgroup comparisons are descriptive, not causal effects of CRAS.

Table 23: Descriptive Quota Responses Across CRAS Adoption Cohorts

	Program Responses		Electoral Outcomes		
	Registration	Enrollment	Δ Incumbent's Vote Share	Δ Reelection Probability	Δ Turnout
	(1)	(2)	(3)	(4)	(5)
Panel A: No CRAS Record in the 2012 Census					
ΔQuota_m (ρ_1)	36.291	40.515**	0.196	5.090	-0.410*
	(16.889)	(10.652)	(1.457)	(5.955)	(0.150)
95% CI	[1.508, 71.075]	[18.578, 62.453]	[-2.805, 3.196]	[-7.175, 17.355]	[-0.719, -0.101]
Number of Mayors	62	62	62	62	62
Observations	3,834	3,834	3,834	3,834	3,834
Panel B: Opened by 2006					
ΔQuota_m (ρ_1)	38.063***	56.645***	1.907***	2.847	-0.299
	(10.542)	(7.810)	(0.634)	(2.452)	(0.264)
95% CI	[16.352, 59.775]	[40.561, 72.729]	[0.602, 3.212]	[-2.203, 7.897]	[-0.842, 0.243]
Number of Mayors	831	831	831	831	831
Observations	3,834	3,834	3,834	3,834	3,834
Panel C: Opened 2007–2008					
ΔQuota_m (ρ_1)	71.237***	82.138***	1.141**	0.264	-0.467***
	(17.532)	(17.347)	(0.497)	(1.943)	(0.181)
95% CI	[35.129, 107.344]	[46.412, 117.865]	[0.117, 2.166]	[-3.738, 4.265]	[-0.840, -0.095]
Number of Mayors	563	563	563	563	563
Observations	3,834	3,834	3,834	3,834	3,834
Panel D: Opened 2009–2010					
ΔQuota_m (ρ_1)	56.968***	65.963***	-0.909	2.053	-0.260*
	(9.309)	(10.211)	(0.893)	(2.286)	(0.116)
95% CI	[37.794, 76.141]	[44.933, 86.993]	[-2.749, 0.930]	[-2.655, 6.762]	[-0.499, -0.022]
Number of Mayors	314	314	314	314	314
Observations	3,834	3,834	3,834	3,834	3,834
Panel E: Opened 2011–September 2012					
ΔQuota_m (ρ_1)	107.150**	85.808***	0.134	4.017	0.113
	(23.862)	(11.408)	(0.562)	(2.261)	(0.156)
95% CI	[58.005, 156.294]	[62.314, 109.302]	[-1.024, 1.292]	[-0.641, 8.674]	[-0.209, 0.435]
Number of Mayors	147	147	147	147	147
Observations	3,834	3,834	3,834	3,834	3,834
Bolsa Família Characteristics	✓	✓	✓	✓	✓
Party Characteristics	✓	✓	✓	✓	✓
Municipality Characteristics	✓	✓	✓	✓	✓
Mayor Characteristics	✓	✓	✓	✓	✓

Notes: All columns absorb municipality and year fixed effects. All columns use 2006-population weights and state-clustered standard errors. The quota instrument and beneficiary-family expansion are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. *Bolsa Família* characteristics include baseline 2008 program intensity, per-family benefit value, and log 2006 quota. Party characteristics include 2008 incumbent party and 2006 PT presidential vote share. Municipality characteristics include baseline 2008 FPM transfers. Predetermined controls are interacted with post. Standard errors are in parentheses. The five cohort slopes are from fully interacted reduced-form models on the common 1,917-mayor (3,834-observation) sample; Number of Mayors shows cohort support, while Observations shows the full jointly estimated sample. Registration and enrollment are rates per 10,000 residents; electoral outcomes are percentage points. Pointwise 95% intervals use the state-clustered covariance and a t_{G-1} critical value. Stars denote null-imposed state wild-cluster bootstrap p-values from 9,999 draws: $*\ p < 0.10$, $**\ p < 0.05$, $***\ p < 0.01$. This test is unavailable for the separately reconstructed reelection column, which therefore receives no markers. Later cohorts use dates reported by offices observed in 2012 and condition on post-reform choices, so these comparisons are descriptive. No 2012 census record does not mean verified absence of an office. Reelection probability is conditional on the incumbent rerunning in 2012. Contrasts and adjusted tests remain in the analysis sidecars.

Table 24: Alternative Exclusion Channels

	Alternative Exclusion Channels	
	(1)	(2)
Panel A: Δ FPM Transfers		
ΔQuota_m	0.082	0.107***
	(0.058)	(0.020)
Panel B: Δ Per-Family *Bolsa Família* Benefit Value		
ΔQuota_m	0.049	0.083*
	(0.043)	(0.035)
Bolsa Família Characteristics	✓	✓
Party Characteristics	✓	✓
Municipality Characteristics		✓
Mayor Characteristics		✓
Number of Mayors	2,078	2,078
Observations	4,156	4,156

Notes: Panel A uses the change between the 12-month average of total municipal FPM transfers in 2008 and 2012, per 10,000 residents using fixed 2006 population, divided by its full-sample 2006-population-weighted SD. Panel B uses the change between the simple averages of the twelve monthly per-family benefit values in 2008 and 2012, divided by the cross-municipality SD of its change. The analysis-weighted mean changes are R$57,959.64 per month per 10,000 residents for FPM and R$49.48 per family per month for benefit value. Multiplying the standardized Column (2) quota coefficients by the corresponding outcome SD gives R$5,466.59 per month per 10,000 residents and R$0.91 per family per month, respectively. All models use 2006-population weights and absorb municipality and year effects. Each displayed row is a reduced-form channel diagnostic and is not conditioned on realized beneficiary expansion. Column (1) includes incumbent-party effects and baseline *Bolsa Família* intensity. Column (2) adds municipality and incumbent-mayor characteristics. These controls are baseline and pre-shock: party, program intensity, and mayor characteristics use 2008 values; population uses 2006; and the remaining municipality characteristics use the 2000 Census. Both columns therefore use only pre-shock controls. Standard errors clustered by state are reported in parentheses. Stars use state wild-cluster bootstrap p-values: $* \; p < 0.10$, $* * \; p < 0.05$, $* * * \; p < 0.01$.

Table 25: Where the Electoral Response Appears: Reduced Forms

	Main (1)	w/o CRAS (2)	w/ CRAS (3)	Δ (4)
Panel A: Incumbent Votes				
ΔQuota_m (ρ_1)	0.950* (0.375)	-0.141 (0.454)	1.253** (0.425)	1.395** (0.509)
Panel B: Other Candidates' Votes				
ΔQuota_m (ρ_1)	-0.875** (0.316)	0.312 (0.470)	-1.169*** (0.381)	-1.481* (0.641)
Panel C: Blank or Null Ballots				
ΔQuota_m (ρ_1)	-0.403** (0.196)	-0.289 (0.173)	-0.438* (0.255)	-0.149 (0.287)
Panel D: Abstention				
ΔQuota_m (ρ_1)	0.328* (0.196)	0.118 (0.081)	0.353 (0.229)	0.235 (0.212)
Panel E: Registered Voters				
ΔQuota_m (ρ_1)	110.975*** (39.011)	92.051*** (26.219)	117.231*** (45.281)	25.180 (39.992)
Bolsa Família Characteristics	✓	✓	✓	✓
Party Characteristics	✓	✓	✓	✓
Municipality Characteristics	✓	✓	✓	✓
Mayor Characteristics	✓	✓	✓	✓
Number of Mayors	2,078	684	1,394	2,078
Observations	4,156	1,368	2,788	4,156

Notes: All columns absorb municipality and year fixed effects. All columns use 2006-population weights and state-clustered standard errors. The quota instrument and beneficiary-family expansion are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. *Bolsa Família* characteristics include baseline 2008 program intensity, per-family benefit value, and log 2006 quota. Party characteristics include 2008 incumbent party and 2006 PT presidential vote share. Municipality characteristics include baseline 2008 FPM transfers. Predetermined controls are interacted with post. Standard errors are in parentheses. Reduced-form coefficients are responses to one fixed headline-sample weighted standard deviation of quota exposure. The Δ column is w/ CRAS minus w/o CRAS from a fully interacted model, with its own state-clustered standard error. Stars on reduced-form estimates, including Δ, use two-sided null-imposed state wild-cluster bootstrap p-values from 9,999 draws (unadjusted across outcomes): $*\ p < 0.10$, $**\ p < 0.05$, $***\ p < 0.01$. The first four outcomes use registered voters as their denominator and sum to 100; their coefficients sum to zero. The last is registered voters per 10,000 residents. The sample rows count municipalities in each displayed subgroup; the reduced-form subgroup coefficients and difference are estimated jointly on the full sample. Aggregate accounting cannot identify individual voter transitions.

Table 26: Where the Electoral Response Appears: Two-Stage Least Squares

	Main (1)	w/o CRAS (2)	w/ CRAS (3)	Δ (4)
Panel A: Incumbent Votes				
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	2.773† (1.242)	-0.401 (1.303)	3.640† (1.515)	4.042 (1.604)
AR 95% CI	[0.49, 5.96]	[-3.24, 2.41]	[1.00, 7.90]	
Panel B: Other Candidates' Votes				
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	-2.553† (1.061)	0.889 (1.364)	-3.394† (1.311)	-4.282 (1.939)
AR 95% CI	[-5.29, -0.61]	[-1.91, 4.00]	[-6.98, -1.05]	
Panel C: Blank or Null Ballots				
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	-1.176† (0.633)	-0.823 (0.540)	-1.271 (0.841)	-0.449 (0.892)
AR 95% CI	[-2.79, -0.00]	[-2.17, 0.19]	[-3.59, 0.22]	
Panel D: Abstention				
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	0.956 (0.610)	0.335 (0.234)	1.025 (0.709)	0.689 (0.666)
AR 95% CI	[-0.21, 2.46]	[-0.15, 0.86]	[-0.33, 2.84]	
Panel E: Registered Voters				
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	323.905† (94.931)	261.788† (68.274)	340.442† (106.931)	78.654 (103.423)
AR 95% CI	[106.93, 520.20]	[116.02, 411.70]	[88.10, 562.27]	
Kleibergen–Paap F	41.386	55.787	31.756	
Bolsa Família Characteristics	✓	✓	✓	✓
Party Characteristics	✓	✓	✓	✓
Municipality Characteristics	✓	✓	✓	✓
Mayor Characteristics	✓	✓	✓	✓
Number of Mayors	2,078	684	1,394	2,078
Observations	4,156	1,368	2,788	4,156

Notes: All columns absorb municipality and year fixed effects. All columns use 2006-population weights and state-clustered standard errors. The quota instrument and beneficiary-family expansion are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. *Bolsa Família* characteristics include baseline 2008 program intensity, per-family benefit value, and log 2006 quota. Party characteristics include 2008 incumbent party and 2006 PT presidential vote share. Municipality characteristics include baseline 2008 FPM transfers. Predetermined controls are interacted with post. Standard errors are in parentheses. IV coefficients are percentage points per one fixed headline-sample weighted standard deviation of beneficiary-family expansion. The Δ coefficient is With CRAS minus Without CRAS from one fully interacted dual-IV system; its conventional state-clustered standard error uses the covariance between subgroup coefficients. No weak-IV-robust AR confidence set or scalar first-stage F is reported for this contrast. The Kleibergen–Paap F values in the first three columns are just-identified excluded-instrument cluster-robust Wald statistics. Anderson–Rubin 95% confidence sets for those separate fits use quadratic state-clustered inversion and are robust to arbitrary instrument weakness. Disconnected two-tail sets use their conservative full-real-line envelope; exact components remain in the sidecars. The first four outcomes use registered voters as their denominator and sum to 100; their coefficients sum to zero. The last is registered voters per 10,000 residents. The sample rows count municipalities in each displayed subgroup; the IV difference is estimated jointly on the full sample. Aggregate accounting cannot identify individual voter transitions.

Table 27: Reverse Causality Channels

	Reverse Causality Channels	
	(1)	(2)
Panel A: Δ Permitted *Bolsa Família* Expansion		
Incumbent's Vote Share in 2008	0.009	-0.005
	(0.018)	(0.016)
Panel B: Δ Realized *Bolsa Família* Expansion		
Incumbent's Vote Share in 2008	-0.058**	-0.049*
	(0.026)	(0.022)
Bolsa Família Characteristics	✓	✓
Party Characteristics	✓	✓
Municipality Characteristics		✓
Mayor Characteristics		✓
Number of Mayors	2,078	2,078
Observations	2,078	2,078

Notes: Panel A measures permitted quota expansion in quota-shock-SD units. Panel B measures the 2008–2012 net change in the 12-month average beneficiary-family stock in treatment-change-SD units. The regressor is the 2008 winner's vote share divided by its analysis-weighted SD of 9.59 percentage points. Every column is weighted by 2006 population and includes state fixed effects. Column (1) includes incumbent-party effects and baseline *Bolsa Família* intensity. Column (2) adds municipality and incumbent-mayor characteristics. These controls are baseline and pre-shock: party, program intensity, and mayor characteristics use 2008 values; population uses 2006; and the remaining municipality characteristics use the 2000 Census. Both columns therefore use only pre-shock controls. Standard errors clustered by state are reported in parentheses. Stars use state wild-cluster bootstrap p-values: $* \; p < 0.10$, $* * \; p < 0.05$, $* * * \; p < 0.01$.

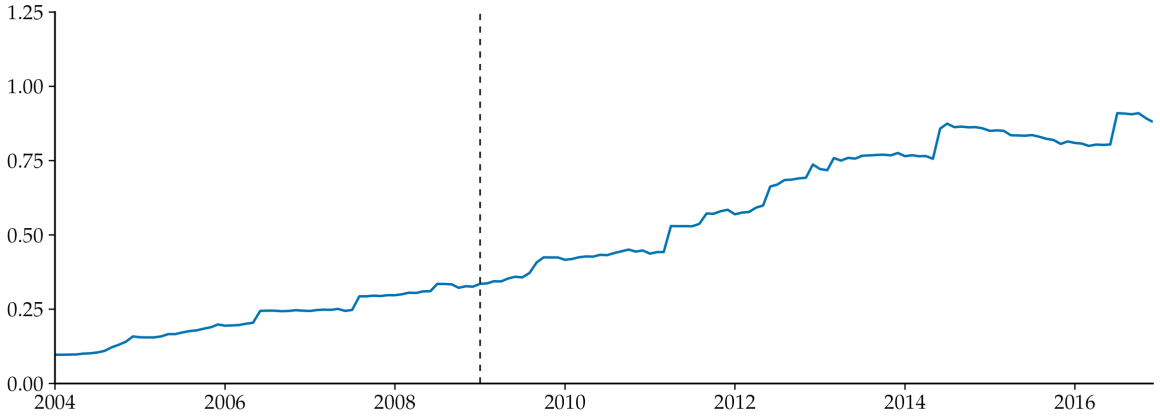
Table 28: Sensitivity by Quota-Shock Size

	ΔBeneficiaries$_m$	
	(1)	(2)
Panel A: Top 50%		
ΔQuota$_m$	0.504***	0.545***
	(0.062)	(0.057)
Panel B: Top 25%		
ΔQuota$_m$	0.485***	0.518***
	(0.091)	(0.071)
Panel C: Top 15%		
ΔQuota$_m$	0.331	0.440**
	(0.153)	(0.116)
Panel D: Top 5%		
ΔQuota$_m$	0.510	0.604**
	(0.181)	(0.124)
Panel E: Top 1%		
ΔQuota$_m$	0.776	0.801
	(0.163)	(0.135)
Bolsa Família Characteristics	✓	✓
Party Characteristics	✓	✓
Municipality Characteristics		✓
Mayor Characteristics		✓
Number of Mayors	2,078	2,078
Observations	2,078	2,078

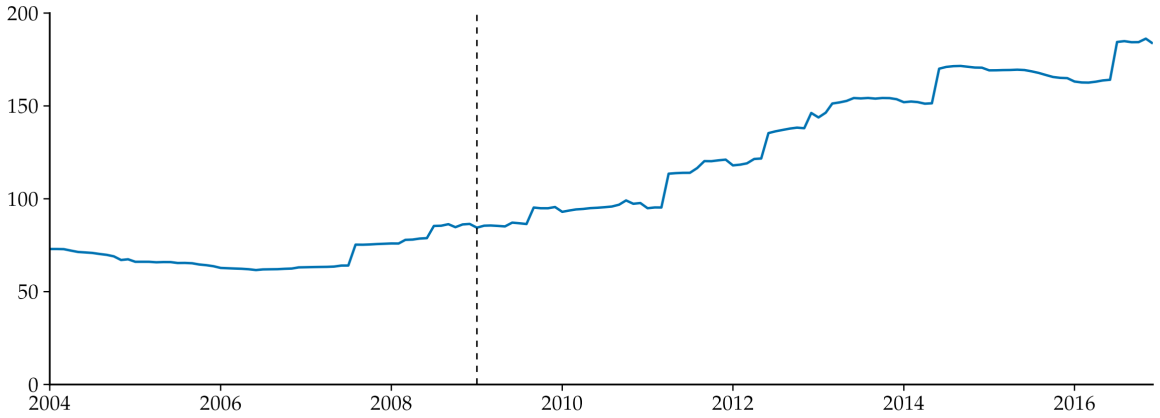
Notes: Each panel estimates separate slopes for the named upper-tail group and its complement on the full locked sample; the table reports the upper-tail slope. Groups are defined by unweighted quota-shock quantiles and are nested. All models include state fixed effects, 2006-population weights, and state-clustered standard errors. Columns (1)–(2) follow the pre-treatment control sequence. Stars use state wild-cluster bootstrap p-values: $*\ p < 0.10$, $**\ p < 0.05$, $***\ p < 0.01$. This is an average first-stage-shape diagnostic, not a test of unit-level monotonicity.

Figure 6: Monthly *Bolsa Família* Payments

Panel A: Monthly Payments (BRL billions)



Panel B: Monthly Payments per Family (BRL)



Notes: Totals cover the same 2,078 headline municipalities throughout, rather than Brazil as a whole. Payments are nominal BRL; the mean benefit is aggregate payments divided by beneficiary families. The series retains the existing data-construction corrections for isolated payment spikes. The dashed line marks the year 2009, not the implementation month. Source: MDS monthly municipal program records, distinct from the 2012 family-registration snapshot.

Table 29: Core Expansion and Family Registrations by CRAS Presence

	Δ Registrations and Other Program Responses			
	Main	w/o CRAS	w/ CRAS	Δ
	(1)	(2)	(3)	(4)
Δ Registrations				
Panel A: All Families				
ΔQuota_m (ρ_1)	48.471***	55.151***	45.860***	-9.291
	(9.465)	(11.873)	(11.314)	(15.037)
Panel B: Income Above R$140				
ΔQuota_m (ρ_1)	4.870	5.406	3.461	-1.945
	(3.649)	(6.488)	(3.964)	(6.104)
Panel C: Income at or Below R$140				
ΔQuota_m (ρ_1)	43.601***	49.744***	42.398***	-7.346
	(8.116)	(7.435)	(9.936)	(11.777)
Panel D: Income at or Below R$70				
ΔQuota_m (ρ_1)	34.323***	39.573***	33.434***	-6.139
	(7.119)	(6.637)	(8.274)	(8.536)
Other Program Responses				
Panel E: Δ Beneficiary Families				
ΔQuota_m (ρ_1)	62.138***	63.772***	62.453***	-1.320
	(9.659)	(8.463)	(11.110)	(10.784)
Panel F: Δ Monthly Payments per Resident				
ΔQuota_m (ρ_1)	0.993***	0.998***	0.993***	-0.005
	(0.155)	(0.131)	(0.180)	(0.172)
Bolsa Família Characteristics	✓	✓	✓	✓
Party Characteristics	✓	✓	✓	✓
Municipality Characteristics	✓	✓	✓	✓
Mayor Characteristics	✓	✓	✓	✓
Number of Mayors	2078	684	1394	2078
Observations	4,156	4,156	4,156	4,156

Notes: All columns absorb municipality and year fixed effects. All columns use 2006-population weights and state-clustered standard errors. The quota instrument and beneficiary-family expansion are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. *Bolsa Família* characteristics include baseline 2008 program intensity, per-family benefit value, and log 2006 quota. Party characteristics include 2008 incumbent party and 2006 PT presidential vote share. Municipality characteristics include baseline 2008 FPM transfers. Predetermined controls are interacted with post. Standard errors are in parentheses. Coefficients report responses to one fixed headline-sample weighted SD of quota exposure. The top header and Panels A–D report Δ registrations; Panels E–F report complementary program responses. First registrations count families first registered in 2009–2012 that survive in the 2012 snapshot; beneficiary families measure the change in the 12-month average stock receiving *Bolsa Família*; monthly payments per resident measure the transfer flow. These are distinct outcomes, not sequentially verified states of the same families. Registration and beneficiary-family quantities are per 10,000 fixed 2006 residents; payments are nominal monthly BRL per resident. Income is classified in the 2012 snapshot. Stars use Holm-adjusted null-imposed state wild-cluster bootstrap p-values across the six outcomes in each column, using 9,999 draws: $*\ p < 0.10$, $**\ p < 0.05$, $***\ p < 0.01$. The Main column fits a common-slope specification on the displayed analysis sample. The Δ column is w/ minus w/o from the interacted specification. Number of Mayors counts each subgroup; Observations gives the full jointly estimated sample. CRAS is measured before the reform.

Table 30: Core Expansion and Family Registrations by Active Outreach

	Δ Registrations and Other Program Responses			
	Main	w/o Outreach	w/ Outreach	Δ
	(1)	(2)	(3)	(4)
Δ Registrations				
Panel A: All Families				
ΔQuota_m (ρ_1)	45.860***	69.017**	40.603***	-28.414
	(11.286)	(21.262)	(7.985)	(18.374)
Panel B: Income Above R$140				
ΔQuota_m (ρ_1)	3.461	4.326	3.601	-0.725
	(3.954)	(4.074)	(4.590)	(4.943)
Panel C: Income at or Below R$140				
ΔQuota_m (ρ_1)	42.398***	64.691***	37.001***	-27.690
	(9.911)	(19.312)	(6.273)	(16.265)
Panel D: Income at or Below R$70				
ΔQuota_m (ρ_1)	33.434***	51.751***	29.097***	-22.654
	(8.253)	(15.712)	(5.615)	(12.875)
Other Program Responses				
Panel E: Δ Beneficiary Families				
ΔQuota_m (ρ_1)	62.453***	86.830***	57.098***	-29.731
	(11.083)	(19.754)	(7.603)	(15.331)
Panel F: Δ Monthly Payments per Resident				
ΔQuota_m (ρ_1)	0.993***	1.365***	0.910***	-0.455
	(0.179)	(0.308)	(0.131)	(0.239)
Bolsa Família Characteristics	✓	✓	✓	✓
Party Characteristics	✓	✓	✓	✓
Municipality Characteristics	✓	✓	✓	✓
Mayor Characteristics	✓	✓	✓	✓
Number of Mayors	1394	377	1017	1394
Observations	2,788	2,788	2,788	2,788

Notes: All columns absorb municipality and year fixed effects. All columns use 2006-population weights and state-clustered standard errors. The quota instrument and beneficiary-family expansion are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. *Bolsa Família* characteristics include baseline 2008 program intensity, per-family benefit value, and log 2006 quota. Party characteristics include 2008 incumbent party and 2006 PT presidential vote share. Municipality characteristics include baseline 2008 FPM transfers. Predetermined controls are interacted with post. Standard errors are in parentheses. Coefficients report responses to one fixed headline-sample weighted SD of quota exposure. The top header and Panels A–D report Δ registrations; Panels E–F report complementary program responses. First registrations count families first registered in 2009–2012 that survive in the 2012 snapshot; beneficiary families measure the change in the 12-month average stock receiving *Bolsa Família*; monthly payments per resident measure the transfer flow. These are distinct outcomes, not sequentially verified states of the same families. Registration and beneficiary-family quantities are per 10,000 fixed 2006 residents; payments are nominal monthly BRL per resident. Income is classified in the 2012 snapshot. Stars use Holm-adjusted null-imposed state wild-cluster bootstrap p-values across the six outcomes in each column, using 9,999 draws: $*\ p < 0.10$, $**\ p < 0.05$, $***\ p < 0.01$. The Main column fits a common-slope specification on the displayed analysis sample. The Δ column is w/ minus w/o from the interacted specification. Number of Mayors counts each subgroup; Observations gives the full jointly estimated sample. Outreach uses verified yes/no baseline reports within CRAS municipalities.

Table 31: Core Expansion and Family Registrations by PAIF Family Follow-Up

	Δ Registrations and Other Program Responses			
	Main (1)	w/o PAIF (2)	w/ PAIF (3)	Δ (4)
Δ Registrations				
Panel A: All Families				
ΔQuota_m (ρ_1)	45.860***	53.048**	43.867***	-9.180
	(11.286)	(21.237)	(9.737)	(19.139)
Panel B: Income Above R$140				
ΔQuota_m (ρ_1)	3.461	1.172	4.618	3.446
	(3.954)	(4.462)	(4.691)	(5.690)
Panel C: Income at or Below R$140				
ΔQuota_m (ρ_1)	42.398***	51.876**	39.249***	-12.627
	(9.911)	(20.693)	(8.189)	(18.203)
Panel D: Income at or Below R$70				
ΔQuota_m (ρ_1)	33.434***	38.318*	31.526***	-6.791
	(8.253)	(18.547)	(6.921)	(16.796)
Other Program Responses				
Panel E: Δ Beneficiary Families				
ΔQuota_m (ρ_1)	62.453***	74.248**	60.806***	-13.442
	(11.083)	(24.049)	(9.416)	(21.747)
Panel F: Δ Monthly Payments per Resident				
ΔQuota_m (ρ_1)	0.993***	1.142***	0.972***	-0.170
	(0.179)	(0.378)	(0.146)	(0.338)
Bolsa Família Characteristics	✓	✓	✓	✓
Party Characteristics	✓	✓	✓	✓
Municipality Characteristics	✓	✓	✓	✓
Mayor Characteristics	✓	✓	✓	✓
Number of Mayors	1394	432	962	1394
Observations	2,788	2,788	2,788	2,788

Notes: All columns absorb municipality and year fixed effects. All columns use 2006-population weights and state-clustered standard errors. The quota instrument and beneficiary-family expansion are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. *Bolsa Família* characteristics include baseline 2008 program intensity, per-family benefit value, and log 2006 quota. Party characteristics include 2008 incumbent party and 2006 PT presidential vote share. Municipality characteristics include baseline 2008 FPM transfers. Predetermined controls are interacted with post. Standard errors are in parentheses. Coefficients report responses to one fixed headline-sample weighted SD of quota exposure. The top header and Panels A–D report Δ registrations; Panels E–F report complementary program responses. First registrations count families first registered in 2009–2012 that survive in the 2012 snapshot; beneficiary families measure the change in the 12-month average stock receiving *Bolsa Família*; monthly payments per resident measure the transfer flow. These are distinct outcomes, not sequentially verified states of the same families. Registration and beneficiary-family quantities are per 10,000 fixed 2006 residents; payments are nominal monthly BRL per resident. Income is classified in the 2012 snapshot. Stars use Holm-adjusted null-imposed state wild-cluster bootstrap p-values across the six outcomes in each column, using 9,999 draws: $*\ p < 0.10$, $**\ p < 0.05$, $***\ p < 0.01$. The Main column fits a common-slope specification on the displayed analysis sample. The Δ column is w/ minus w/o from the interacted specification. Number of Mayors counts each subgroup; Observations gives the full jointly estimated sample. PAIF is the existing baseline recorded-family indicator within CRAS municipalities; missing counts retain the canonical zero coding. This proxy is not a verified contact-frequency measure.

Figure 7: Annual Registration Responses by Baseline CRAS



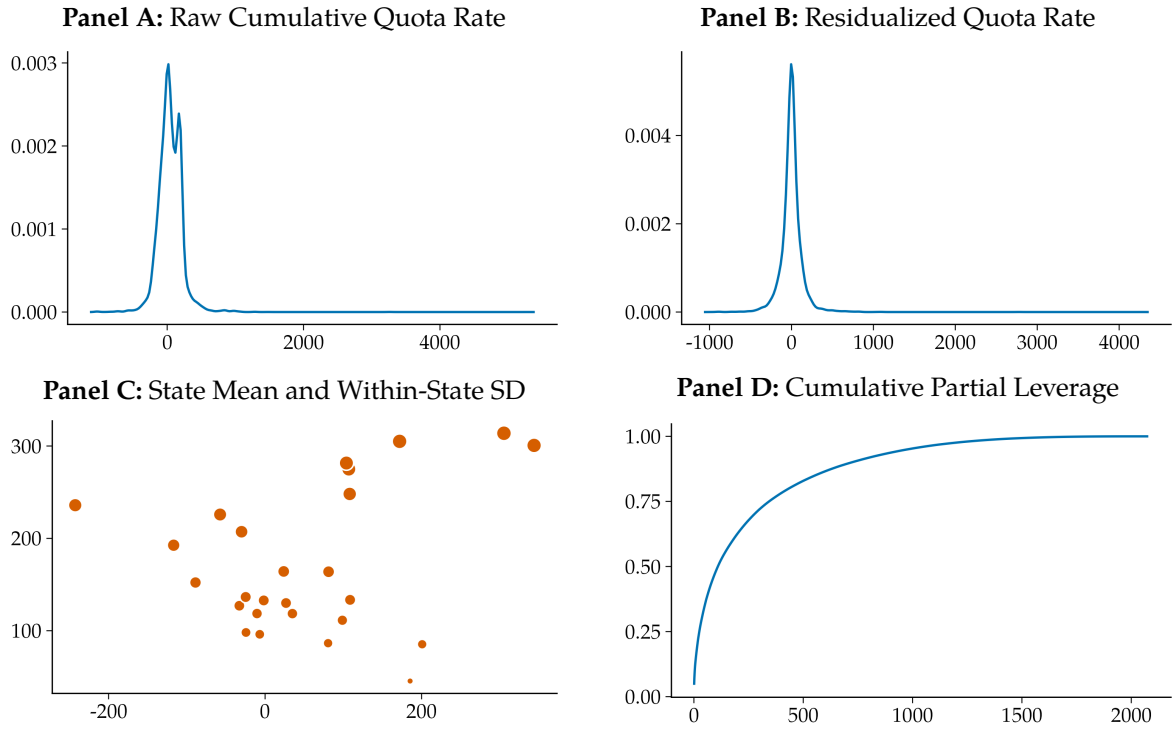
Notes: Each point is the quota-related response of the indicated registration cohort, per 10,000 residents, for municipalities with or without a recorded CRAS before the reform. The two panels use separate authorization measures, each scaled by its own full-sample population-weighted standard deviation. Bars are pointwise state-clustered 95% confidence intervals. Table F.19 reports the group contrasts and multiplicity-adjusted bootstrap tests. Families are observed retrospectively in the 2012 snapshot. Panel B decomposes the cumulative main response by registration year; it does not imply that a future quota change caused earlier registrations. The available aggregates cannot identify pre-2009 monthly trends or families that exited the registry.

Table 32: Electoral Returns and Household Transfers: Conditional Registration Scenarios

	Estimated	Assumed Registration-Dependent Share			
	Quota Response	0%	25%	50%	100%
	(1)	(2)	(3)	(4)	(5)
Panel A: Family Registrations					
Response / Assumed Reduction	48.47***	48.47	48.47	48.47	48.47
	(9.47)				
95% CI	[28.98, 67.97]	[28.98, 67.97]	[28.98, 67.97]	[28.98, 67.97]	[28.98, 67.97]
Panel B: Beneficiary Families					
Response / Assumed Reduction	62.14***	0.00	15.53	31.07	62.14
	(9.66)				
95% CI	[42.25, 82.03]	[0.00, 0.00]	[10.56, 20.51]	[21.12, 41.02]	[42.25, 82.03]
Panel C: Monthly Transfers (BRL)					
Response / Assumed Reduction	9,929.11***	0.00	2,482.28	4,964.56	9,929.11
	(1,549.96)				
95% CI	[6,736.91, 13,121.31]	[0.00, 0.00]	[1,684.23, 3,280.33]	[3,368.45, 6,560.66]	[6,736.91, 13,121.31]
Panel D: Incumbent Vote Share					
Response / Assumed Reduction	1.06**	0.00	0.26	0.53	1.06
	(0.38)				
95% CI	[0.27, 1.84]	[0.00, 0.00]	[0.07, 0.46]	[0.14, 0.92]	[0.27, 1.84]
Number of Mayors	2,078				
Observations	4,156				

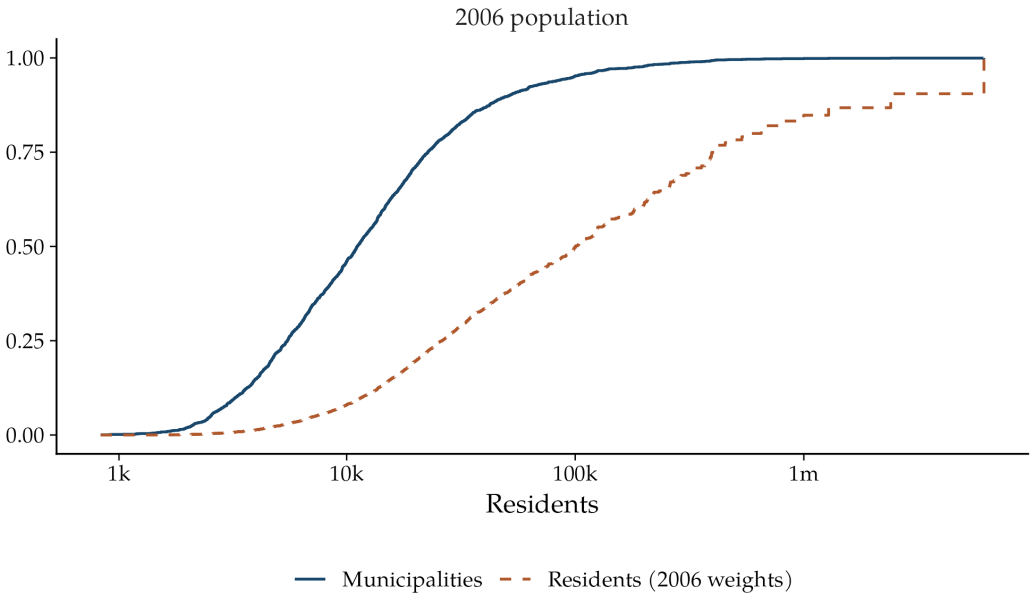
Notes: Column (1) estimates responses to 171.50 additional quota slots per 10,000 residents, using full baseline controls, municipality/year effects, 2006-population weights and state-clustered standard errors (parentheses). Intervals are pointwise conventional 95% intervals, not Anderson–Rubin sets. Stars on estimated responses use null-imposed state wild-cluster bootstrap p-values after Holm adjustment across the four outcomes (9,999 draws): $*\ p < 0.10$, $**\ p < 0.05$, $***\ p < 0.01$; scenario values are assumptions, not estimates. Registrations are surviving 2009–2012 entry cohorts in the 2012 snapshot; beneficiary counts and monthly payments use annual averages. Panels A–C are per 10,000 residents; payments are nominal BRL and votes are percentage points. Columns (2)–(5) suppress all quota-induced registrations (Panel A). For each other outcome, the assumed share multiplies column (1); matching shares across outcomes are illustrative, not estimated. Zero permits gains through other channels, including previously registered families; 100% assumes all gains depend on new registrations. Conditional intervals treat these shares as fixed. These scenarios do not identify mediation or the effect of removing political incentives. Transfers are municipal totals, not payments linked to new registrants, total household income, cumulative transfers or welfare.

Figure 8: Quota Support, Within-State Variation, and Partial Leverage



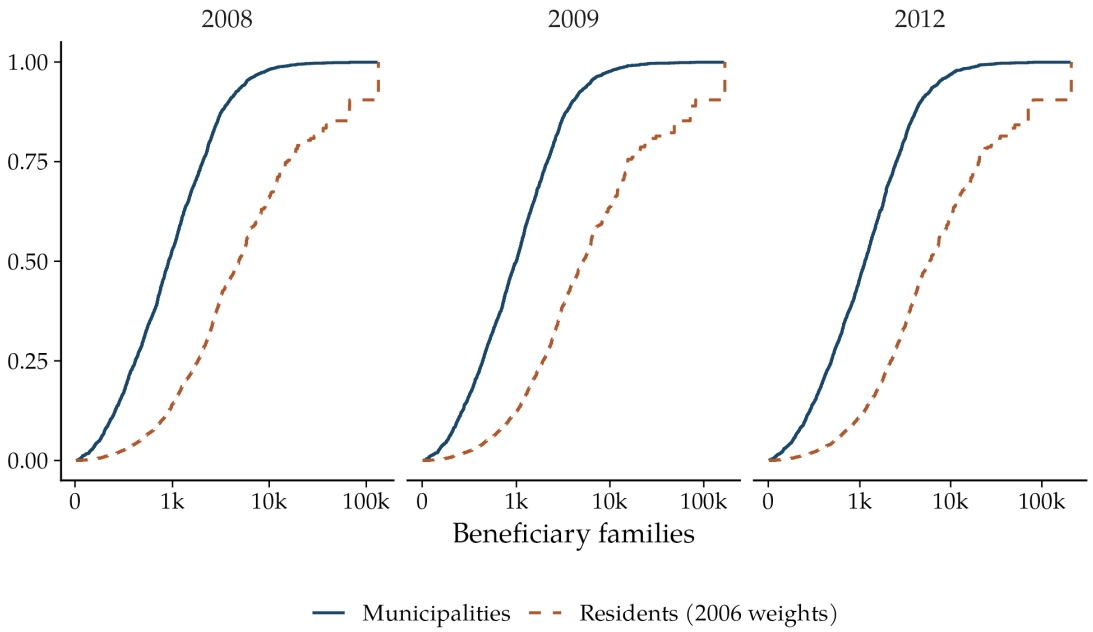
Notes: The locked headline sample has 2,078 municipalities and uses 2006-population weights. Panel B partials out the full predetermined headline controls, incumbent-party indicators, and state indicators. Partial leverage is proportional to weighted squared residual quota exposure. The figure diagnoses concentration and does not define an observation-deletion rule.

Figure 9: Distribution of 2006 Population in the Main Sample



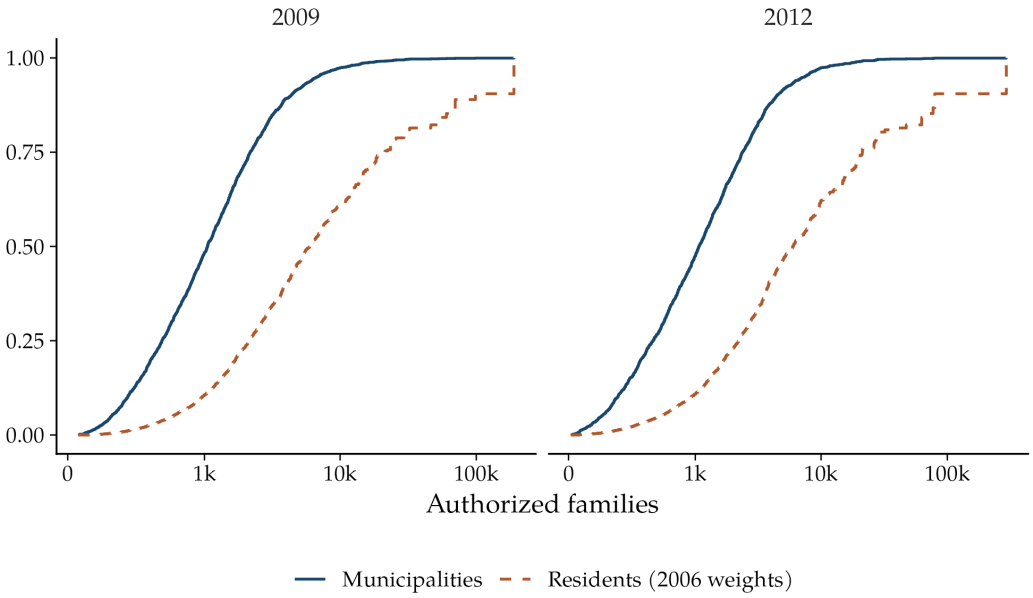
Notes: The sample contains the 2,078 municipalities whose 2008 winners sought reelection in 2012 and meet the complete-case requirements of the main analysis. The solid curve gives each municipality equal weight; the dashed curve weights municipalities by their fixed 2006 population. The horizontal axis uses a pseudo-log scale while labels report resident counts. These are descriptive cumulative distributions, not estimates of policy effects.

Figure 10: Distribution of Beneficiary-Family Stocks in the Main Sample



Notes: Each panel shows the cumulative distribution of the 12-month average number of families receiving *Bolsa Família* at the December endpoint in the indicated year, for the same 2,078 municipalities. The solid curve weights municipalities equally; the dashed curve uses fixed 2006 population weights. The pseudo-log horizontal scale displays family counts, not a log-rate transformation. Stocks at different endpoints do not by themselves measure program-induced enrollment.

Figure 11: Distribution of Authorized Quotas in the Main Sample



Notes: Each panel shows the federally authorized number of *Bolsa Família* family places at the 2009 or 2012 endpoint in the locked main-analysis municipalities. The solid curve weights municipalities equally; the dashed curve uses fixed 2006 population weights. The pseudo-log horizontal scale displays family counts. These quota stocks are descriptive and differ from the counterfactual-adjusted authorization shock used as the instrument.

Table 33: Electoral Returns Across Single Control Additions

	Main (1)	*Bolsa Família* (2)	Party (3)	Municipality (4)	Mayor (5)
Panel A: First Stage					
ΔQuota_m (π_1)	0.343^{***} (0.053)	0.234^{***} (0.053)	0.309^{***} (0.083)	0.345^{***} (0.062)	0.270^{***} (0.088)
Kleibergen–Paap F	41.4	19.8	14.0	30.7	9.5
Panel B: Reduced Form					
ΔQuota_m (ρ_1)	1.056^{**} (0.381)	2.178^{**} (0.704)	1.941^{**} (0.712)	1.854^{**} (0.632)	1.779^{**} (0.658)
Panel C: Two-Stage Least Squares					
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	$3.084^{\dagger}$ (1.282)	$9.321^{\dagger}$ (3.683)	$6.274^{\dagger}$ (3.316)	$5.370^{\dagger}$ (2.128)	$6.584^{\dagger}$ (3.917)
AR 95% CI	[0.742, 6.400]	[2.958, 20.830]	[1.240, 19.561]	[1.519, 11.158]	[1.161, 28.196]
Bolsa Família Characteristics	✓	✓			
Party Characteristics	✓		✓		
Municipality Characteristics	✓			✓	
Mayor Characteristics	✓				✓
Number of Mayors	2,078	2,078	2,078	2,078	2,078
Observations	4,156	4,156	4,156	4,156	4,156

Notes: All columns absorb municipality and year effects. All columns use 2006-population weights and state-clustered standard errors. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. *Bolsa Família* characteristics comprise baseline 2008 program intensity, per-family benefit value, and log 2006 quota. Party characteristics comprise 2008 incumbent party and 2006 PT presidential vote share. Municipality characteristics include baseline 2008 FPM transfers; mayor characteristics are the predetermined incumbent attributes. Predetermined controls are interacted with post. Standard errors are in parentheses. Reduced-form stars use two-sided null-imposed state wild-cluster bootstrap p-values (9,999 draws): $*\,p < 0.10$, $**\,p < 0.05$, $***\,p < 0.01$. The same star rule applies to first-stage coefficients. † on a 2SLS estimate means its unrounded Anderson–Rubin 95% set excludes zero. Anderson–Rubin 95% confidence sets use quadratic state-clustered inversion with a t_{G-1} critical value and are robust to arbitrary instrument weakness. All columns contain the same 2,078 mayors and 4,156 municipality-year observations. Each nonmain column adds only the named block to the no-control baseline; additions are not cumulative.

Table 34: Electoral Returns Across Cumulative Control Additions

	Baseline (1)	+*Bolsa Família* (2)	+Party (3)	+Municipality (4)	+Mayor (5)
Panel A: First Stage					
ΔQuota_m (π_1)	0.267***	0.234***	0.252***	0.331***	0.343***
	(0.096)	(0.053)	(0.051)	(0.053)	(0.053)
Kleibergen–Paap F	7.7	19.8	24.3	38.3	41.4
Panel B: Reduced Form					
ΔQuota_m (ρ_1)	2.381**	2.178**	1.863**	1.584**	1.056**
	(1.127)	(0.704)	(0.571)	(0.537)	(0.381)
Panel C: Two-Stage Least Squares					
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	8.920†	9.321†	7.395†	4.785†	3.084†
	(6.254)	(3.683)	(2.932)	(1.881)	(1.282)
AR 95% CI	[0.174, 51.220]	[2.958, 20.830]	[2.442, 16.315]	[1.361, 9.711]	[0.742, 6.400]
Bolsa Família Characteristics		✓	✓	✓	✓
Party Characteristics			✓	✓	✓
Municipality Characteristics				✓	✓
Mayor Characteristics					✓
Number of Mayors	2,078	2,078	2,078	2,078	2,078
Observations	4,156	4,156	4,156	4,156	4,156

Notes: All columns absorb municipality and year effects. All columns use 2006-population weights and state-clustered standard errors. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. *Bolsa Família* characteristics comprise baseline 2008 program intensity, per-family benefit value, and log 2006 quota. Party characteristics comprise 2008 incumbent party and 2006 PT presidential vote share. Municipality characteristics include baseline 2008 FPM transfers; mayor characteristics are the predetermined incumbent attributes. Predetermined controls are interacted with post. Standard errors are in parentheses. Reduced-form stars use two-sided null-imposed state wild-cluster bootstrap p-values (9,999 draws): $*\,p < 0.10$, $**\,p < 0.05$, $***\,p < 0.01$. The same star rule applies to first-stage coefficients. † on a 2SLS estimate means its unrounded Anderson–Rubin 95% set excludes zero. Anderson–Rubin 95% confidence sets use quadratic state-clustered inversion with a t_{G-1} critical value and are robust to arbitrary instrument weakness. All columns contain the same 2,078 mayors and 4,156 municipality-year observations. Columns add the four blocks from left to right; the final column is the complete main specification.

Table 35: Electoral Returns Across Control Removals

	Main (1)	− *Bolsa Família* (2)	− Party (3)	− Municipality (4)	− Mayor (5)
Panel A: First Stage					
ΔQuota_m (π_1)	0.343^{***} (0.053)	0.361^{***} (0.062)	0.336^{***} (0.049)	0.255^{***} (0.051)	0.331^{***} (0.053)
Kleibergen–Paap F	41.4	33.7	46.8	24.9	38.3
Panel B: Reduced Form					
ΔQuota_m (ρ_1)	1.056^{**} (0.381)	1.177^{**} (0.380)	1.094^{**} (0.367)	1.368^{***} (0.403)	1.584^{**} (0.537)
Panel C: Two-Stage Least Squares					
$\Delta\widehat{\text{Beneficiaries}}_m$ (β_1)	$3.084^{\dagger}$ (1.282)	$3.265^{\dagger}$ (1.316)	$3.250^{\dagger}$ (1.192)	$5.361^{\dagger}$ (2.242)	$4.785^{\dagger}$ (1.881)
AR 95% CI	[0.742, 6.400]	[0.979, 6.939]	[0.984, 6.174]	[1.752, 12.484]	[1.361, 9.711]
Bolsa Família Characteristics	✓		✓	✓	✓
Party Characteristics	✓	✓		✓	✓
Municipality Characteristics	✓	✓	✓		✓
Mayor Characteristics	✓	✓	✓	✓	
Number of Mayors	2,078	2,078	2,078	2,078	2,078
Observations	4,156	4,156	4,156	4,156	4,156

Notes: All columns absorb municipality and year effects. All columns use 2006-population weights and state-clustered standard errors. The instrument and endogenous variable are rates per 10,000 residents using fixed 2006 population and are reported per one analysis-weighted SD. *Bolsa Família* characteristics comprise baseline 2008 program intensity, per-family benefit value, and log 2006 quota. Party characteristics comprise 2008 incumbent party and 2006 PT presidential vote share. Municipality characteristics include baseline 2008 FPM transfers; mayor characteristics are the predetermined incumbent attributes. Predetermined controls are interacted with post. Standard errors are in parentheses. Reduced-form stars use two-sided null-imposed state wild-cluster bootstrap p-values (9,999 draws): $*\,p < 0.10$, $**\,p < 0.05$, $***\,p < 0.01$. The same star rule applies to first-stage coefficients. † on a 2SLS estimate means its unrounded Anderson–Rubin 95% set excludes zero. Anderson–Rubin 95% confidence sets use quadratic state-clustered inversion with a t_{G-1} critical value and are robust to arbitrary instrument weakness. All columns contain the same 2,078 mayors and 4,156 municipality-year observations. Each nonmain column omits only the named block from the complete main specification.